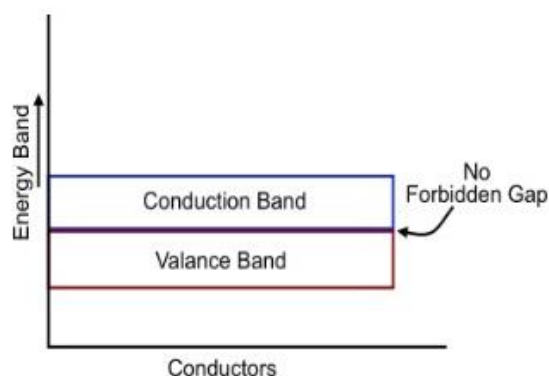


MODULE 2

1. **Conductors** are the substance or material that transmit heat or electricity through them. Such elements or materials that may act as a conductor, as well as insulators under different conditions, are known as semiconductors, whereas **Insulators** are the substance or material that does not transmit heat or electricity to through them.
2. Conductors have **high conductivity**, while semiconductors have moderate and insulators have **low (negligible) conductivity**.
3. **Forbidden gap** (the gap between the valence band and conduction band) is not found in conductors, while there is a small forbidden gap in semiconductors and the large forbidden gap is seen in insulators.
4. **Resistivity** (the property of the resistance to measure the electrical conduction), which is low or negligible in conductors and very high in insulators, while moderate in semiconductors.
5. **Conductivity value** (higher is the conductivity value, the lower is the substance resistivity) is 10^{-7}mho/m (very high) of the conductors, whereas semiconductors have the value between 10^{-7}mho/m to 10^{-13}mho/m and insulators have 10^{-13}mho/m (negligible).
6. Resistivity value of the conductors is less than $10^{-5}\ \Omega\text{-m}$, while the semiconductors hold the value between $10^{-5}\ \Omega\text{-m}$ to $10^5\ \Omega\text{-m}$ and insulators is having more than $10^5\ \Omega\text{-m}$.
7. There is only one valence electron in the outermost shell in conductors, though there are four valence electrons in the outermost shell of the semiconductors and there are eight valence electrons in the outermost shell of the insulators.
8. Conductors are formed by metallic bonding; Semiconductors are formed by covalent bonding; Insulators are formed by ionic bonds.

Conductor

A conductor is a type of material that allows the electric current to flow through it i.e., it possesses least resistance in the path of free electrons. In case of conductor, the valance and conduction bands overlap. Due to this overlapping, a small potential difference across a conductor causes the free electrons to constitute electric current.

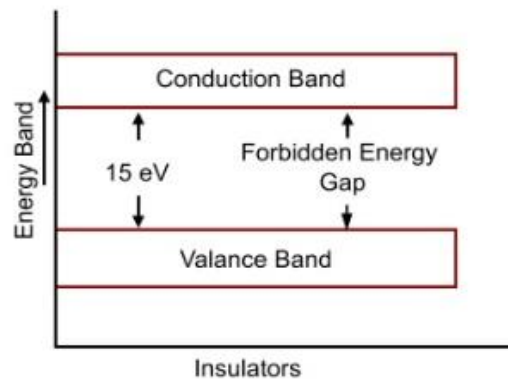


All the metals are conductors. The resistance of the conductors increases with the increase in the temperature. Hence, the conductor have ***positive temperature co-efficient of resistance***.

The conductors are used for making conductor wires and cables for carrying electric current.

Insulator

An insulator is type of material that does not allow the electric current to pass through it, due to its high electrical resistance. In the insulators, the energy gap between valance and conduction bands is very large (***about 15 eV***). Therefore, a very high electric field is required to push the valance electrons to the conduction band. Due to this, there are no free electrons in the conduction band. For this reason, the electrical conductivity of insulators is very low and may considered nil under ordinary conditions.

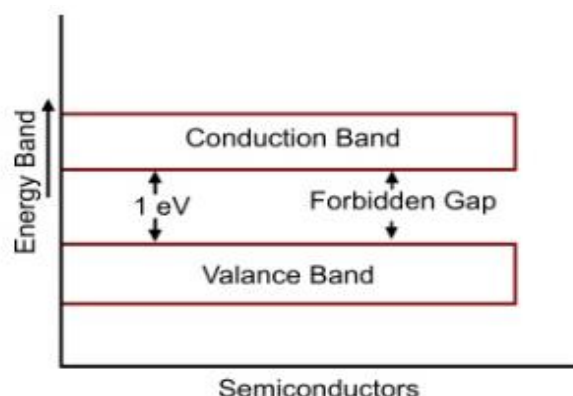


At room temperature, the valance electrons of the insulator do not have enough energy to cross over to the forbidden energy gap. But, if the temperature is raised, some of the electrons may acquire enough energy to cross over to the forbidden energy gap. Hence, the resistance of the insulator decreases with the rise in temperature. Therefore, the insulators have ***negative temperature co-efficient of resistance***.

Due to high electrical resistance, the insulators are used for protection against electric shocks.

Semiconductor

The semiconductors are the materials having conductivity in-between conductors and insulators. In a semiconductor, the forbidden energy gap between valance and conduction bands is very small (***about 1 eV***) as compared to insulators. Therefore, a smaller electric field (smaller than insulators but greater than conductors) is required to push the free electrons from valance band to the conduction band.



At low temperature, the valance band of semiconductor is completely full and the conduction band is completely empty. Thus, a semiconductor behaves as an insulator at low temperature. However, at room temperature, some electrons can cross the forbidden energy gap, imparting a little conductivity to the semiconductor. As temperature is increased, more valance electrons cross over to the energy gap to reach to the conduction band and the conductivity increases. This shows that electrical conductivity of semiconductor increases with the rise in temperature. Hence, a semiconductor has ***negative temperature coefficient of resistance***.

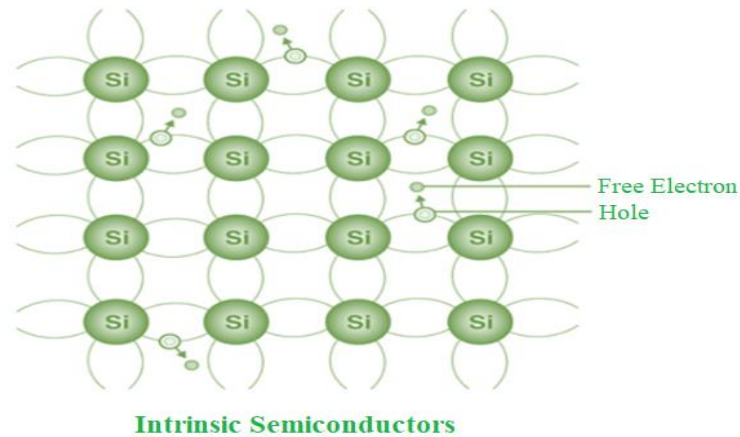
The conductivity of semiconductors can also be increased by adding some impurity in the pure semiconductor material, called ***doping***. The semiconductors are commonly used in manufacturing of solid state electronic devices.

Difference between Conductor, Semiconductor and Insulator

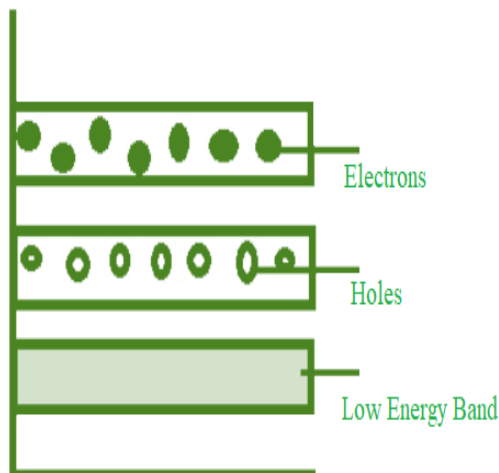
Parameter	Conductor	Semiconductor	Insulator
Definition	A material that allows electric current to pass through it very easily.	A material that has conductivity in between conductors and insulators.	Materials that do not allow the electric current to pass through them.
Forbidden Energy Gap	No energy gap i.e., the conduction band overlap the valance band.	Small energy gap (approx. 1 eV).	Very large energy gap (approx. 15 eV).
Conductivity	High Conductivity (of the order of 10^{-7} mho/m).	Intermediate conductivity (ranging from 10^{-7} mho/m to 10^{-13} mho/m).	Very low conductivity(of the order of 10^{-13} mho/m).
Conduction	Due to free electrons.	Due to movement of both electrons and holes (positive charge carriers).	No conduction.
Resistivity	Low (of the order of $10^{-5}\Omega/\text{m}$).	Intermediate (from $10^{-5}\Omega/\text{m}$ to $10^5 \Omega/\text{m}$).	Very high (of the order of $10^5 \Omega/\text{m}$).
Temperature Coefficient of Resistivity	Positive	Negative	Negative
Valance Electrons in Outermost Shell	Less than 4	4	More than 4
Examples	Metals like silver (best conductor), gold, copper, aluminium etc.	Silicon, Germanium, Gallium, Arsenide etc.	Air, Mica, Glass, Paper, Porcelain, Wood etc.
Application	In the manufacturing of conducting wires and cables.	In the manufacturing of solid state electronic devices like ICs, diodes, transistors etc.	Used for providing insulation electrical and electronic devices, for preventing electric shock etc.

Intrinsic Semiconductor

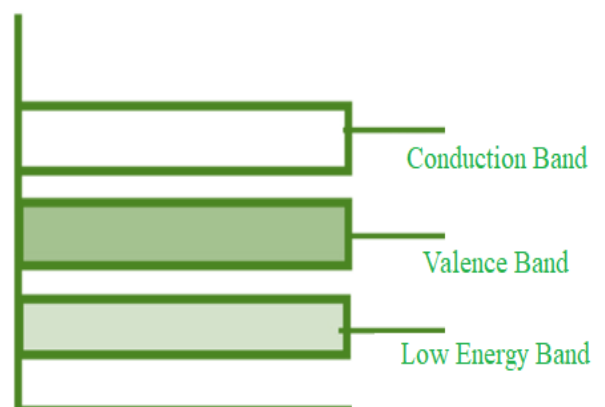
The definition of an intrinsic semiconductor is a semiconductor that is exceedingly pure. According to the energy band theory, the conductivity of this semiconductor will be zero at ambient temperature. Si and Ge are two examples of intrinsic semiconductors.



- The conduction band is empty in the below energy band diagram, but the valence band is completely filled. Some heat energy can be provided to it once the temperature has been raised. As a result of exiting the valence band, electrons from the valence band are provided to the conduction band.



Energy Band in
Intrinsic Semiconductors



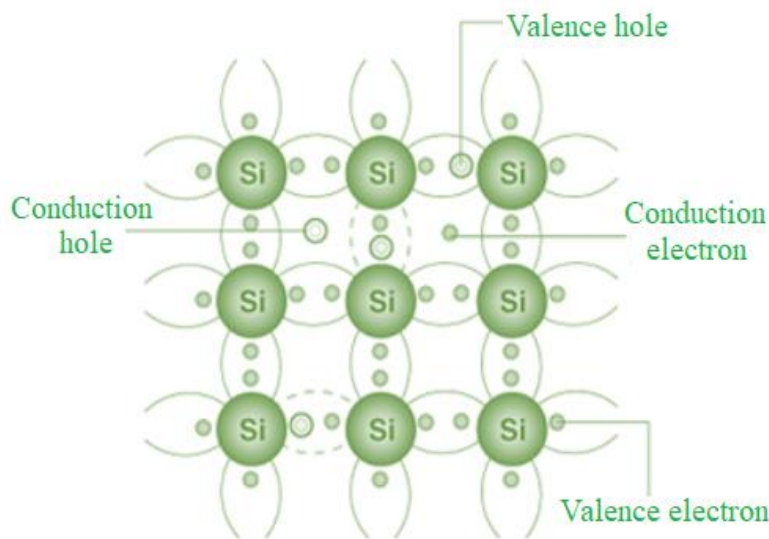
Energy Band in
Intrinsic Semiconductors

- The flow of electrons will be random as they go from the valence to the conduction band. The crystal's holes can also flow freely in any direction.
- As a result, the TCR of this semiconductor will be negative (temperature coefficient of resistance). The TCR indicates that when the temperature rises, the material's resistance decreases and its conductivity rises.

Extrinsic Semiconductor

Extrinsic semiconductors are semiconductors that have had an impurity introduced to them at a regulated rate to make them conductive.

- While insulating materials may be doped to make them into semiconductors, intrinsic semiconductors can also be doped to make an extrinsic semiconductor.
- Extrinsic semiconductors are divided into two categories as a result of doping: atoms with an additional electron (n-type for negative, from group V) and atoms with one fewer electron (p-type for positive, from group III).
- Doping is the purposeful introduction of impurities into a very pure, or intrinsic, semiconductor in order to change its electrical characteristics. The kind of semiconductor determines the impurities. Extrinsic semiconductors are those that are light to moderately



doped.

What is Doping?

Doping is the process of introducing an impurity into a semiconductor. During the production of extrinsic semiconductors, the amount and kind of impurity to be introduced to the material must be carefully monitored. In most cases, one impurity atom is introduced to every 10^8 semiconductor atoms.

Impurity is used to enhance the number of free electrons or holes in a semiconductor crystal, making it more conductive. A significant number of free electrons will exist if a pentavalent impurity with five valence electrons is introduced to a pure semiconductor. A significant number of holes will exist in the semiconductor if a trivalent impurity with three valence electrons is introduced. Extrinsic semiconductors are divided into two categories based on the type of impurity added: N-type and P-type semiconductors.

n-Type Semiconductors

N-type semiconductors are extrinsic semiconductors in which dopant atoms can provide additional conduction electrons to the host material (e.g. phosphorus in silicon).

An overabundance of negative (n-type) electron charge carriers results as a result of this. Doping atoms often contain one extra valence electron than the host atoms. Atomic replacement in group-IV

solids by group-V elements is the most typical case. When the host includes many types of atoms, the issue becomes more complicated. Silicon, for example, can act as a donor when it replaces gallium or as an acceptor when it replaces arsenic in III-V semiconductors like gallium arsenide. Some donors have fewer valence electrons than the host, such as alkali metals, which are donors in most solids.

p-Type Semiconductors

To enhance the number of free charge carriers, a p-type (p for “positive”) semiconductor is formed by adding a certain type of atom to the semiconductor.

The doping substance removes (accepts) weakly bonded outer electrons from semiconductor atoms when it is introduced. The vacancy left behind by the electron is known as a hole, and this sort of doping agent is also known as an acceptor substance. The goal of p-type doping is to produce a large number of holes.

In the instance of silicon, the crystal lattice is swapped by a trivalent atom. As a result, one of the four covalent bonds that typically make up the silicon lattice is lacking an electron. As a result, the dopant atom can accept an electron from the covalent link of a nearby atom to complete the fourth bond. Acceptors are the name given to these dopants because of this.

When a dopant atom takes an electron, it causes the nearby atom to lose half of its link, resulting in the creation of a hole. Each hole is connected to a neighbouring negatively charged dopant ion, resulting in an electrically neutral semiconductor. Once each hole has strayed away into the lattice, one proton in the atom at the hole’s position will be “exposed,” meaning it will no longer be cancelled by an electron. This atom will contain three electrons and one hole in its nucleus, which will have four protons.

As a result, a hole behaves like a positive charge. When a large enough number of acceptor atoms are supplied, thermally excited electrons are substantially outnumbered by holes. In p-type materials, holes are the majority carriers, whereas electrons are the minority carriers.

Difference between Intrinsic and Extrinsic Semiconductors

The following are some of the key distinctions between extrinsic and intrinsic semiconductors:

- Intrinsic semiconductors exist in their purest form at all times while Extrinsic semiconductors are created by doping impurities in pure semiconductors.
- At room temperature, Intrinsic semiconductors have poor electrical conductivity while Extrinsic semiconductors have a high electrical conductivity compared to other materials.
- The number of electrons equals the number of holes in Intrinsic semiconductors while numbers are unequal in Extrinsic semiconductors.
- Intrinsic semiconductors are solely reliant on temperature while Extrinsic semiconductors are affected by temperature and the number of contaminants present.
- Intrinsic semiconductors are not further classified while N-type and p-type semiconductors are two types of semiconductors in Extrinsic semiconductors.
- Silicon and germanium are two examples of Intrinsic semiconductors while Si and Ge doped with Al, In, P, As, and other elements are examples of Extrinsic semiconductors
-

Parameter	Intrinsic Semiconductor	Extrinsic Semiconductor
Definition	The semiconductor in its pure form is known as intrinsic semiconductor.	When a chemical impurity is added to an intrinsic semiconductor, then the resulting semiconductor is known as extrinsic semiconductor.
Types	There is no classification of intrinsic semiconductor.	Based on the impurity added, the extrinsic semiconductors are of two types viz. P-type semiconductor and N-type semiconductor.
Doping	In case of intrinsic semiconductor, there is no doping or addition of impurity.	In an extrinsic semiconductor, the doping is performed, i.e. a small amount of impurity is added in the pure semiconductor.
Charge carrier density	In an intrinsic semiconductor, the number of electrons is equal to number of holes.	In case of extrinsic semiconductors, the number of holes and electrons are not equal. In a P-type semiconductor, the holes are more than electrons while in an N-type semiconductor, the electrons are more than holes.
Conductivity	The electrical conductivity of intrinsic semiconductors is low.	The extrinsic semiconductors have high electrical conductivity.
Dependency of conductivity	The conductivity of intrinsic semiconductor is dependent on temperature only.	The conductivity of extrinsic semiconductor is dependent on temperature as well as amount of impurity added.
Conductivity at 0 Kelvin	The intrinsic semiconductor does not conduct at 0 Kelvin temperature.	The extrinsic semiconductor conducts at 0 Kelvin temperature.
Cause of charge carrier production	In the intrinsic semiconductors, the charge carriers are produced only due to thermal agitation.	In case of extrinsic semiconductors, the charge carriers are produced due to thermal agitation as well as chemical impurities added to the semiconductor.
Operating temperature	Intrinsic semiconductors have low operating temperature.	The operating temperature for extrinsic semiconductors is high.
Fermi level	At 0 Kelvin, the Fermi level in intrinsic semiconductors exactly lies between conduction band and valance band.	At 0 Kelvin, the Fermi level in extrinsic semiconductor depends upon the types of semiconductor, i.e. in case of N-type semiconductor, it lies closer to conduction band and in the P-type semiconductor, it lies closer to valance band.
Ratio of charge carriers	In an intrinsic semiconductor, the ratio of majority and minority charge carriers is 1.	In case of extrinsic semiconductor, the ratio of majority and minority charge carriers is not unity.
Examples	The crystalline forms of pure silicon and germanium are the examples of intrinsic semiconductors.	The examples of extrinsic semiconductors are pure silicon and germanium doped with chemical impurities like As, P, Bi, Sb, In, B, Al, etc

What Is Fermi Energy?

Named after the Physicist Enrico Fermi, a Fermi level is the measure of the energy of the least tightly held electrons within a solid. It is important in determining the thermal and electrical properties of solids. It can be defined as:

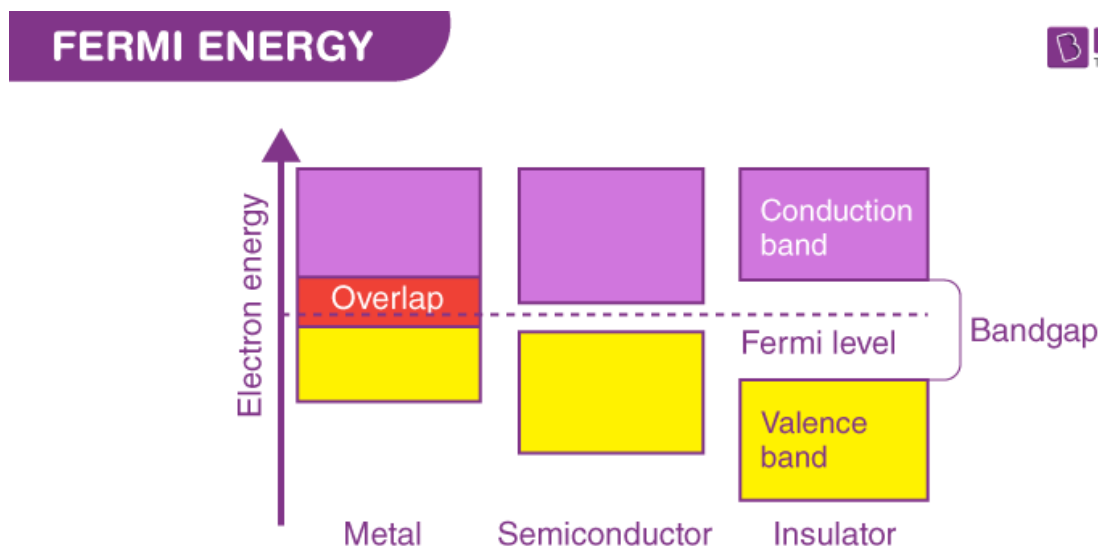
Fermi energy is a concept in quantum mechanics that usually refers to the energy difference between the highest and lowest occupied single-particle states in a quantum system of non-interacting fermions at absolute zero temperature.

The value of the Fermi level at absolute zero temperature ($-273.15\text{ }^{\circ}\text{C}$) is known as the **Fermi energy**. It is also the maximum kinetic energy an electron can attain at 0K. Fermi energy is constant for each solid.

Calculating Fermi Energy

To determine the lowest possible Fermi energy of a system, we first group the states with equal energy into sets and arrange them in increasing order of energy. We then add particles one at a time, successively filling up the unoccupied quantum states with the lowest energy.

When all the particles are arranged accordingly, the energy of the highest occupied state is the Fermi energy. In spite of the extraction of all possible energy from metal by cooling it to near absolute zero temperature (0 Kelvin), the electrons in the metal still move around. The fastest ones move at a velocity corresponding to a kinetic energy equal to the Fermi energy.



The Fermi level and Fermi energy are usually confusing terms and are often used interchangeably to refer to each other. Although both the terms are equal at absolute zero temperature, they are different at other temperatures.

Fermi energy is applied in determining the electrical and thermal characteristics of the solids. It is one of the important concepts in superconductor physics and quantum mechanics. It is used in semiconductors and insulators.

What Is Fermi Level?

The highest energy level that an electron can occupy at the absolute zero temperature is known as the **Fermi Level**. The Fermi level lies between the valence band and conduction band because at absolute zero temperature, the electrons are all in the lowest energy state. Due to the lack of sufficient energy at 0 Kelvin, the Fermi level can be considered as the sea of fermions (or electrons) above which no electrons exist. The Fermi level changes as the solids are warmed and as electrons are added to or withdrawn from the solid.

Fermi Energy Level

The reason for the existence of this energy level is due to Pauli's exclusion principle, which states two fermions cannot occupy that same quantum state. So, if a system has more than one fermion, each fermion has a different set of magnetic quantum numbers associated with it.

The Fermi Temperature can be defined as the energy of the Fermi level divided by the Boltzmann’s constant. It is also the temperature at which the energy of the electron is equal to the Fermi energy. It is the measure of the electrons in the lower states of energy in metal.

Applications of Fermi Energy

It is one of the important concepts in quantum mechanics and condensed matter physics. Some Fermi energy applications are given in the points below.

- It is used in semiconductors and insulators.
- It is used to describe insulators, metals, and semiconductors.
- Fermi energy is applied in determining the electrical and thermal characteristics of the solids.
- It is also important in nuclear physics to understand the stability of white dwarfs. White dwarfs are stars that have a mass comparable to the Sun but have about a hundredth of its radius.

Value of Fermi energy for different elements

Element	Fermi Energy eV
Li	4.74
K	2.12
Na	3.24
Cs	1.59
Rb	1.85
Ag	5.49
Cu	7.00
Be	14.3
Au	5.53
Ca	4.69
Mg	7.08
Ba	3.64
Sr	3.93

Fe	11.1
Nb	5.32
Zn	9.47
Mn	10.9
Hg	7.13
Cd	7.47
Al	11.7
Ga	10.4
In	8.63
Tl	8.15
Sn	10.2
Pb	9.47
Bi	9.90
Sb	10.9

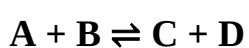
What is the Law of Mass Action?

The law of mass action states that the rate of a reaction is proportional to the product of the concentrations of each reactant.

This law can be used to explain the behavior exhibited by solutions in dynamic equilibria. The law of mass action also suggests that the ratio of the reactant concentration and the product concentration is constant at a state of chemical equilibrium.

The Equilibrium Constant (K_c)

The concentration of reactants and products, at equilibrium, are constant at a given temperature. Consider the following simple reversible reaction where A & B are the reactants whereas C & D are the products.



A mixture of products and reactants in a state of chemical equilibrium is known as an equilibrium mixture. There exists a relation between the concentration of products and the concentration of reactants for an equilibrium mixture. This relation can be equated as follows.

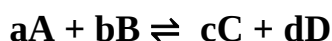
$$K_c = \frac{[C][D]}{[A][B]}$$

Here, K_c is called the equilibrium constant. In this equation, the concentration of A at equilibrium is represented as [A] (similarly for B, C, and D), and the stoichiometric coefficients of the reactants and products are 1. It has been experimentally observed that the equilibrium constant is also dependent on the stoichiometric coefficients of the reactants and products.

Therefore, the law of mass action dictates that the equilibrium constant, at a given constant temperature, is equal to the product of the concentration of products raised to the respective stoichiometric coefficients divided by the product of the reactant concentrations, each raised to the corresponding stoichiometric coefficient.

Representation of the Equilibrium Constant

For a balanced reaction of the type,



According to the law of mass action, the constant value obtained by relating equilibrium concentrations of reactants and products is called the equilibrium constant. For the forward reaction, this is given by

$$K_c = \frac{[C]^c [D]^d}{[A]^a [B]^b}$$

The equilibrium constant for the reverse reaction is the inverse of the forward reaction and is given by:

$$K_c' = \frac{1}{K_c} = \frac{[A]^a [B]^b}{[C]^c [D]^d}$$

If the coefficients of the chemical equation are multiplied by a factor 'n' then the equilibrium constant is raised by the power 'n' i.e. the constant becomes K_c^n .

Equilibrium Constant Representation	Expressed in terms of	Expressed as
K_c	Concentrations of reactants and products	$\frac{[C]^c [D]^d}{[A]^a [B]^b}$
K_p	Partial pressures of reactants and products. (only for the substances which are in gaseous state)	$\frac{p_C^c p_D^d}{p_A^a p_B^b}$
K_x	Mole fractions of reactants and products	$\frac{[X_C]^c [X_D]^d}{[X_A]^a [X_B]^b}$

Relation between K_c , K_p and K_x

$$K_p = K_c(RT)^{\Delta n_g}$$

$$K_p = K_x \cdot P^{\Delta n_g}$$

Where,

Δn_g = moles of gaseous products – moles of gaseous reactants

K_c is the equilibrium constant expressed in terms of the concentration of the reactants/products. Similarly, K_p is the constant in terms of the partial pressures of the substances and K_x is expressed in terms of the mole fraction.

Applications of the Law of Mass Action

This law is also applicable to semiconductors and, therefore, has several important implications in the fields of electronics and semiconductor physics. Here, the law of mass action provides a relationship between the concentrations of electron holes and free electrons when the semiconductor system is in a state of thermal equilibrium.

The law of mass action also has applications in the following fields:

- Mathematical ecology
- Sociophysics (social physics)
- Mathematical epidemiology

FAQs

1. What is mass law constant?

Ans: Mass action law states that a chemical reaction frequency is proportional to the active masses of reacting materials at a constant temperature. K_c is the reaction constant of equilibrium, while K_p is the constant of equilibrium found by applying partial pressure.

2. What are K_p and K_c ?

Ans: K_c and K_p are the constants of the equilibrium of gas mixtures. The distinction between the two constants, however, is that K_c is determined by molar concentrations, while K_p is defined by the gasses' partial pressures within a closed system.

3. What is the law of mass action examples?

Ans: Law of mass action, the law stating that the frequency of any chemical reaction is proportional to the sum of the masses of the reacting materials, each mass being elevated to a power equal to the coefficient in the chemical equation.

4. How the active mass is represented?

Ans: Usually active mass in units of mol dm^{-3} is known to be the molar concentration; expressed as square brackets. [].

5. What is an effective concentration in chemistry?

Ans: Activity is a measure of a species' active concentration under conditions that are not optimal (e.g., concentrated). This determines, rather than an idea, the real chemical potential for a real solution.

Current components in semiconductor

We shall now consider in detail the mechanism of current flow when a voltage is applied to:

1. n-type semiconductor and
2. p-type semiconductor.

1. n-Type Semiconductor:

An n-type semiconductor has – (i) electrons as majority carriers, (ii) an equal number of immobile positive ions and (iii) holes as minority carriers. Assume hole contribution to current is negligible.

Consider an n-type semiconductor placed between a pair of electrodes across which a voltage is applied. Due to the field produced by the voltage there will be steady drift of the free electrons towards the +ve electrode. The electrons reaching the +ve terminal disappear at the electrode and the immobile positive ions in the vicinity of the negative electrode remain un-neutralized due to the drift of the free electrons.

These ions immediately attract electrons from the negative electrode. Thus a continuous flow of electrons from one terminal of the voltage source to the other terminal, from the -ve electrode to the semiconductor and from semiconductor to the +ve electrode is determined by the applied voltage and conductivity of the semiconductor which account for the current flow.

In n-type semiconductor the electron densities and hole densities are different so the equation (i) cannot be used as such. Suppose the electron and hole densities respectively be n and p .

∴ The current density $J = (n\mu_e + p\mu_h) E$

and $\sigma = n\mu_e + p\mu_h$

Since hole density is negligibly small, therefore, we have;

$$\sigma = n\mu_e$$

2. p-Type Semiconductor:

A p-type semiconductor has (i) holes as majority carriers, (ii) an equal number of immobile negative ions and (iii) electrons as minority carriers. In this case, the electron current is considered negligible.

When a p-type semiconductor is placed between a pair of electrodes, across which a voltage is applied, there will be a constant drift of the holes towards the -ve electrodes. On reaching the negative electrode they combine with the electrons coming out of the metal of the negative electrode and disappear. At the same time, an equal number of holes are generated near the positive electrode.

This is because as the holes drift away from the positive electrode they leave behind the un-neutralized immobile negative charges. These charges and the positive electrode give rise to an electric field which causes the ionizing electrons to leave the acceptor atoms and come to the

+ve electrode where they are lost. By losing an electron in the above process the acceptor atom attempts to steal an electron from an adjacent bond.

A hole is thus formed. The generation of holes takes place at the rate equal to the rate at which they disappear at the negative electrode. This rate is again determined by the magnitude of the applied voltage and the conductivity of the semiconductor. The current inside the p-type semiconductor is due to the motion of the holes whereas in the external circuit is due to the motion of electrons.

Again in p-type semiconductor, the electron and the hole densities are not equal and so the equation (i) is not valid in p-type semiconductors. Let n and p be electron and hole densities respectively, therefore, we have current density J as-

$$J = (n\mu_e + p\mu_h) E$$

$$\text{and } \sigma = n\mu_e + p\mu_h$$

Since electron density is negligibly small, therefore, we have;

$$\sigma = p\mu_h$$

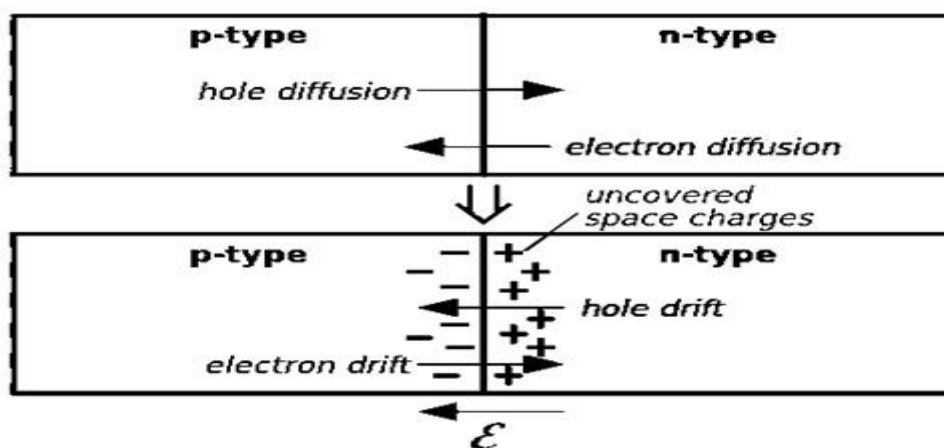
Analysis of Drift and Diffusion Currents:

A semiconductor may be doped with impurity atoms so that the current is due to either electrons or holes in contrast to the metal where current carriers are electrons only. The transportation of the charges in a crystal under the influence of an electric field gives drift current and as a result of non-uniform concentration gradient gives diffusion current.

What is a Drift Current?

Drift current can be defined as the charge carrier's moves in a semiconductor because of the electric field. There are two kinds of charge carriers in a semiconductor like holes and electrons. Once the voltage is applied to a semiconductor, then electrons move toward the +Ve terminal of a battery whereas the holes travel toward the -Ve terminal of a battery.

Here, holes are positively charged carriers whereas the electrons are negatively charged carriers. Therefore, the electrons attract by the +Ve terminal of a battery whereas the holes attract by the -Ve terminal of a battery.



drift-current-&-diffusion-

current

What is Diffusion Current?

The diffusion current can be defined as the flow of charge carriers within a semiconductor travels from a higher concentration region to a lower concentration region. A higher concentration region is nothing but where the number of electrons present in the semiconductor. Similarly, a lower concentration region is where the less number of electrons present in the semiconductor. The process of diffusion mainly occurs when a semiconductor is doped non-uniformly.

In an N-type semiconductor, when it is doped non-uniformly then a higher concentration region can be formed at the left side whereas the lower concentration region can be formed at the right side. The electrons in the higher concentration region are more in the semiconductor so they will experience a repulsive force from each other.

Difference between Drift Current and Diffusion Currents

The difference between drift current and diffusion current includes the following.

Drift Current	Diffusion Current
The movement of charge carriers is because of the applied electric field is known as drift current.	The diffusion current can be occurred because of the diffusion in charge carriers.
It requires electrical energy for the process of drift current.	Some amount of external energy is enough for the process of diffusion current.
This current obeys <u>Ohm's Law</u> .	This current obeys Fick's Law.
The direction of charge carriers in the semiconductor is reverse to each other.	For charge carriers, the densities of diffusion are reverse in symbol to each other.
The direction of the drift current, as well as the electric field, will be the same.	The direction of this current can be decided by the concentration of the carrier slope.
It depends on the permittivity	It is independent of permittivity
The direction of this current mainly depends on the polarity of the applied electric field.	The direction of this current mainly depends on the charge within the concentrations of carrier

FAQs

1). What is drift current in diode?

The charge carriers start moving because of the applied electric field.

2). What is carrier drift?

When an electric field is applied to a semiconductor, charges carriers start moving to generate electric current.

3). What is the drift voltage?

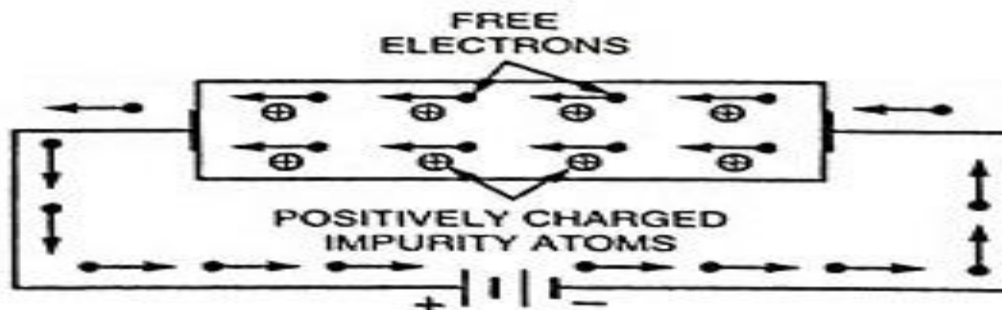
The percentage of o/p voltage occurs over a period of time.

4). What does the diffusion coefficient mean?

The amount of a substance that diffuses from one section to another through every unit of the cross-section for each unit of time as the gradient of volume-concentration is unity.

Continuity Equation in Semiconductor:

The Continuity Equation in Semiconductor states a condition of dynamic equilibrium for the concentration of mobile carriers in any elementary volume of the semiconductor.



In Fig. 6.21 we have seen that on disturbing the equilibrium concentrations of carriers (electrons and holes) in a semiconductor, the concentrations of holes or electrons vary with time approaching the equilibrium value exponentially. In the general case, however, the carrier concentration in the body of a semiconductor is a function of both time and distance. The differential equation governing this functional relationship, called the **Continuity Equation** in Semiconductor, is based upon the fact that charge can be neither created nor destroyed.

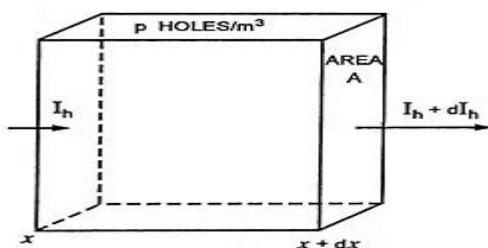


Fig. 6.32

Consider an infinitesimal element of volume of area A and length dx , as shown in Fig. 6.32, within which the average hole concentration is p . If τ_h is the mean lifetime of holes, then p/τ_h equals the holes per second lost by recombination per unit volume. If e is the electronic charge, then, because of recombination, the number of coulombs per second decreases within the volume and

$$\text{Decrease within the volume} = eA dx \frac{p}{\tau_h} \quad \dots(6.108)$$

If g is the thermal rate of generation of electron-hole pairs per unit volume, the number of coulombs per second increases within the volume and

$$\text{Increase within the volume} = eA dx g \quad \dots(6.109)$$

In general, the current varies with distance within the semiconductor. If the current entering the volume at x is I_h and leaving at $x + dx$ is $I_h + dI_h$, as shown in Fig. 6.32, the number of coulombs per second decreases within the volume and

$$\text{Decrease within the volume} = dI_h \quad \dots(6.110)$$

Because of the three effects enumerated above, the hole concentration must change with time, and the total number of coulombs per second increases within the volume and

$$\text{Increase within the volume} = eA dx \frac{dp}{dt} \quad \dots(6.111)$$

Since the charge must be conserved,

$$eA dx \frac{dp}{dt} = -eA dx \frac{p}{\tau_h} + eA dx g - dI_h \quad \dots(6.112)$$

The hole current I_h is the sum of drift current and diffusion current so

$$I_h = AEp e\mu_h - AeD_h \frac{dp}{dx} \quad \dots(6.113)$$

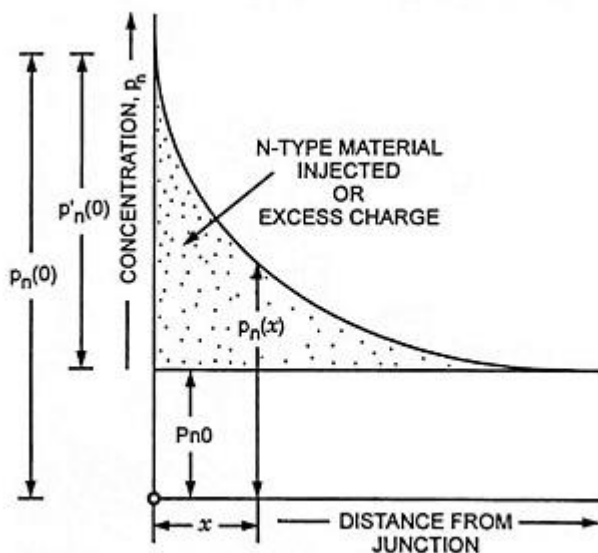


Fig. 6.33 Hole Concentration Components in N-Type Material

If the semiconductor is in thermal equilibrium with its surroundings and is subjected to no applied fields, the hole density will attain a constant value p_0 . Under these conditions, $I_h = 0$ and $dp/dt = 0$. So from the Eq. (6.112), we have

$$g = \frac{p_0}{\tau_h} \quad \dots(6.114)$$

The above equation indicates that the rate of thermal generation of holes equals the rate of holes lost due to recombination, under equilibrium conditions.

Combining Eqs. (6.112), (6.113) and (6.114) we have the equation of conservation of charge, called the **continuity equation**

$$\frac{dp}{dt} = -\frac{p - p_0}{\tau_h} + D_h \frac{d^2 p}{dx^2} - \mu_h \frac{d(pE)}{dx} \quad \dots(6.115)$$

The Eq. (6.115) is the **continuation equation** or **equation of conservation** of charge for holes stating the condition of dynamic equilibrium for the density of mobile carrier holes. Since p is a function of both t and x , partial derivatives should be used and Eq. (6.115) may be modified as

$$\frac{\partial p}{\partial t} = -\frac{p - p_0}{\tau_h} + D_h \frac{\partial^2 p}{\partial x^2} - \mu_h \frac{\partial(pE)}{\partial x} \quad \dots(6.116)$$

Similarly, the Continuity Equation in Semiconductor for electrons states the condition of dynamic equilibrium for the density of mobile carrier electrons and is given by

$$\frac{\partial n}{\partial t} = -\frac{n - n_0}{\tau_e} + D_e \frac{\partial^2 n}{\partial x^2} - \mu_e \frac{\partial(nE)}{\partial x} \quad \dots(6.117)$$

Minority Carrier Injection,

in electronics, a process taking place at the boundary between p-type and n-type semiconductor materials, used in some types of transistors. Each semiconductor material contains two types of freely moving charges: electrons (negative charges) and holes (positive charges). Electrons are the more abundant, or majority, carrier in n -type materials, holes being the less abundant, or minority, carrier. In p -type materials, however, holes are the majority carrier, and electrons the minority carrier. If a battery is properly connected to the semiconductor material, the p -type material may acquire additional electrons (minority carriers), injected into the p -type material from the n -type material by the flow of electrons from the battery. This is minority carrier injection. It is important in bipolar junction transistors, which are made of two p - n junctions.

Hall Effect

Hall effect is a process in which a transverse electric field is developed in a solid material when the material carrying an electric current is placed in a magnetic field that is perpendicular to the current. The Hall effect was discovered by Edwin Herbert Hall in 1879. In this article, let us learn about the Hall effect in detail.

Principle of Hall effect

The principle of the Hall effect states that when a current-carrying conductor or a semiconductor is introduced to a perpendicular magnetic field, a voltage can be measured at the right angle to the current path. This effect of obtaining a measurable voltage is known as the Hall effect.

Theory

When a conductive plate is connected to a circuit with a battery, then a current starts flowing. The charge carriers will follow a linear path from one end of the plate to the other end. The motion of charge carriers results in the production of magnetic fields. When a magnet is placed near the plate, the magnetic field of the charge carriers is distorted. This upsets the straight flow of the charge carriers. The force which upsets the direction of flow of charge carriers is known as Lorentz force.

Due to the distortion in the magnetic field of the charge carriers, the negatively charged electrons will be deflected to one side of the plate and positively charged holes to the other side. A potential difference, known as the Hall voltage will be generated between both sides of the plate which can be measured using a metre.

The Hall voltage represented as V_H is given by the formula:

$$V_H = IBqnd$$

Here,

I is the current flowing through the sensor

B is the magnetic field strength

q is the charge

n is the number of charge carriers per unit volume

d is the thickness of the sensor.

Similar Reading:

<u>Magnetic Field of Electric Current</u>	<u>Magnetic Field</u>
<u>Magnetic Field Due to Current Carrying Conductor</u>	<u>Biot Savart Law</u>

Hall Coefficient

The Hall coefficient R_H is mathematically expressed as

$$R_H = E_j B$$

Where j is the current density of the carrier electron, E_y is the induced electric field and B is the magnetic strength. The hall coefficient is positive if the number of positive charges is more than the negative charges. Similarly, it is negative when electrons are more than holes.

Applications of Hall Effect

Hall effect principle is employed in the following cases:

- Magnetic field sensing equipment
- For the measurement of direct current, Hall effect Tong Tester is used.
- It is used in phase angle measurement
- Proximity detectors

- Hall effect Sensors and Probes
- Linear or Angular displacement transducers
- For detecting wheel speed and accordingly assist the anti-lock braking system.

Name one practical use of Hall effect.

Hall effect is used to determine if a substance is a semiconductor or an insulator. The nature of the charge carriers can be measured.

How is Hall potential developed?

When a current-carrying conductor is in the presence of a transverse magnetic field, the magnetic field exerts a deflecting force in the direction perpendicular to both magnetic field and drift velocity. This causes charges to shift from one surface to another thus creating a potential difference.

What is a Hall effect sensor?

A Hall effect sensor is a device that is used to measure the magnitude of a magnetic field.

In the Hall effect, the direction of the magnetic field and electric field are parallel to each other. True or False?

False. The magnetic field and electric field are perpendicular to each other.

Explain Lorentz Force.

Lorentz force is the force exerted on a charged particle q moving with velocity v through an electric field E and magnetic field B .

Hall Effect Derivation

Consider a metal with one type of charge carrier that is electrons and is a steady-state condition with no movement of charges in the y -axis direction. Following is the derivation of the Hall-effect:

$$eEH = BevV_H d = BevV_H = Bvd$$

(at equilibrium, force is downwards due to magnetic field which is equal to upward electric force)

Where,

- V_H is Hall voltage
- E_H is Hall field
- v is the drift velocity
- d is the width of the metal slab
- B is the magnetic field
- Bev is a force acting on an electron

$$I = -nevA$$

Where,

- I is an electric current
- n is no. of electrons per unit volume
- A is the cross-sectional area of the conductor

$$V_H = -B \frac{I}{net}$$

$$E_H = - \frac{B I}{net}$$

Where,

EHJB: Hall coefficient(R_H) is defined as the ratio between the induced electric field and to the product of applied magnetic field and current density.

In semiconductors, R_H is positive for the hole and negative for free electrons.

$$R_H = -1/ne$$

$$\mu_H = vE = JneE = \sigma R_H = R_H \rho(v)$$

Where,

- E is an electric field
- v is the drift velocity
- R_H is the Hall coefficient
- μ_H is the mobility of the hole

$$J_y J_x = \sigma E_y J_x = \mu_H B_z = \sigma R_H B_z$$

The ratio between density (x-axis direction) and current density (y-axis direction) is known as the Hall angle, which measures the average number of radians due to collisions of the particles.

$$R = V_H i = B_{net}$$

Where,

- R is Hall resistance

Hall Effect Derivation in Semiconductors

In semiconductors, electrons and holes contribute to different concentrations and mobilities, making it difficult to explain the Hall coefficient given above. Therefore, for the simple explanation of a moderate magnetic field, the following is the Hall coefficient:

$$R_H = \frac{p\mu_H^2 - n\mu_e^2}{e(p\mu_H + n\mu_e)}$$

$$\therefore R_H = \frac{(p - nb^2)}{e(p + nb^2)}$$

Where,

$$b = \mu_e \mu_H$$

- n is electron concentration
- p is hole concentration
- μ_e is the mobility of electron
- μ_H is the mobility of the hole
- e is an elementary charge

What is P-N Junction?

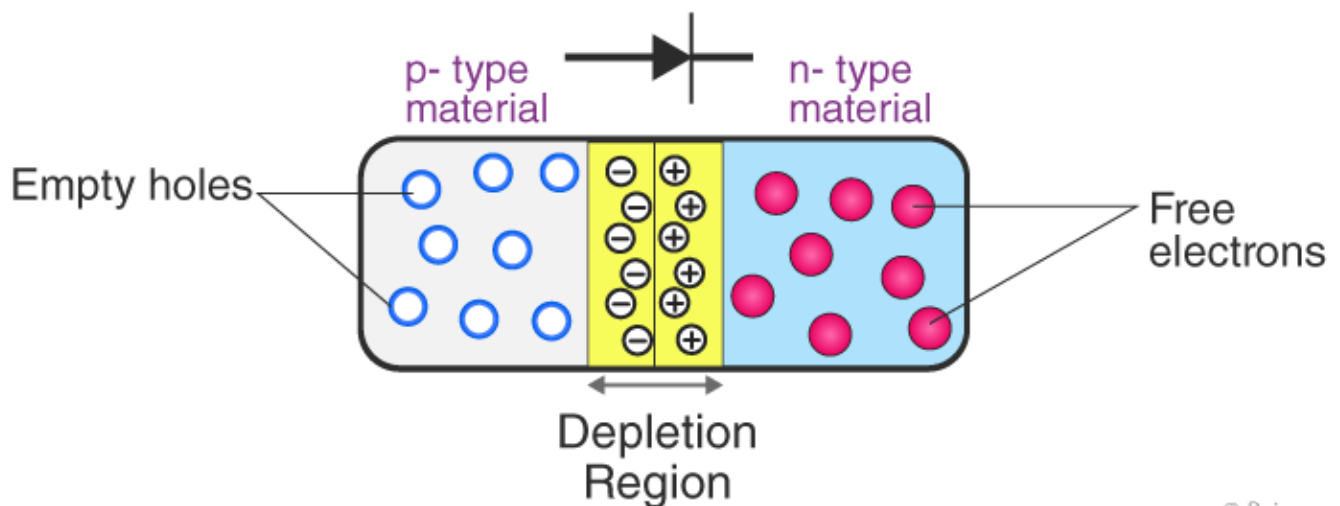
Definition: A P-N junction is an interface or a boundary between two semiconductor material types, namely the p-type and the n-type, inside a semiconductor.

In a semiconductor, the P-N junction is created by the method of doping. The p-side or the positive side of the semiconductor has an excess of holes, and the n-side or the negative side has an excess of electrons. The process of doping is explained in further detail in the next section.

Formation of P-N Junction

As we know, if we use different semiconductor materials to make a P-N junction, there will be a grain boundary that would inhibit the movement of electrons from one side to the other by scattering the electrons and holes and thus, we use the process of doping. We will understand the process of doping with the help of this example. Let us consider a thin p-type silicon semiconductor sheet. If we add a small amount of pentavalent impurity to this, a part of the p-type Si will get converted to n-type silicon. This sheet will now contain both the p-type region and the n-type region and a junction between these two regions. The processes that follow after forming a P-N junction are of two types – diffusion and drift. There is a difference in the concentration of holes and electrons at the two sides of a junction. The holes from the p-side diffuse to the n-side, and the electrons from the n-side diffuse to the p-side. These give rise to a diffusion current across the junction.

UNBIASED P-N JUNCTION



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Also, when an electron diffuses from the n-side to the p-side, an ionised donor is left behind on the n-side, which is immobile. As the process goes on, a layer of positive charge is developed on the n-side of the junction. Similarly, when a hole goes from the p-side to the n-side, an ionized acceptor is left behind on the p-side, resulting in the formation of a layer of negative charges in the p-side of the junction. This region of positive charge and negative charge on either side of the junction is termed as the depletion region. Due to this positive space charge region on either side of the junction, an electric field with the direction from a positive charge towards the negative charge is developed. Due to this electric field, an electron on the p-side of the junction moves to the n-side of the junction. This motion is termed the drift. Here, we see that the direction of the drift current is opposite to that of the diffusion current.

Biasing Conditions for the P-N Junction Diode

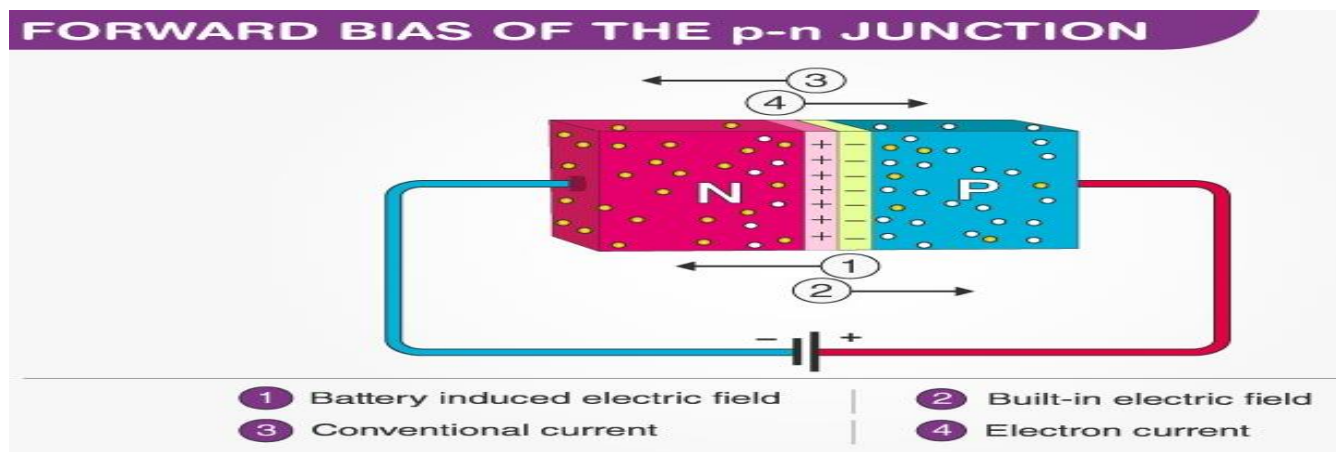
There are two operating regions in the P-N junction diode:

- P-type
- N-type

There are three biasing conditions for the P-N junction diode, and this is based on the voltage applied:

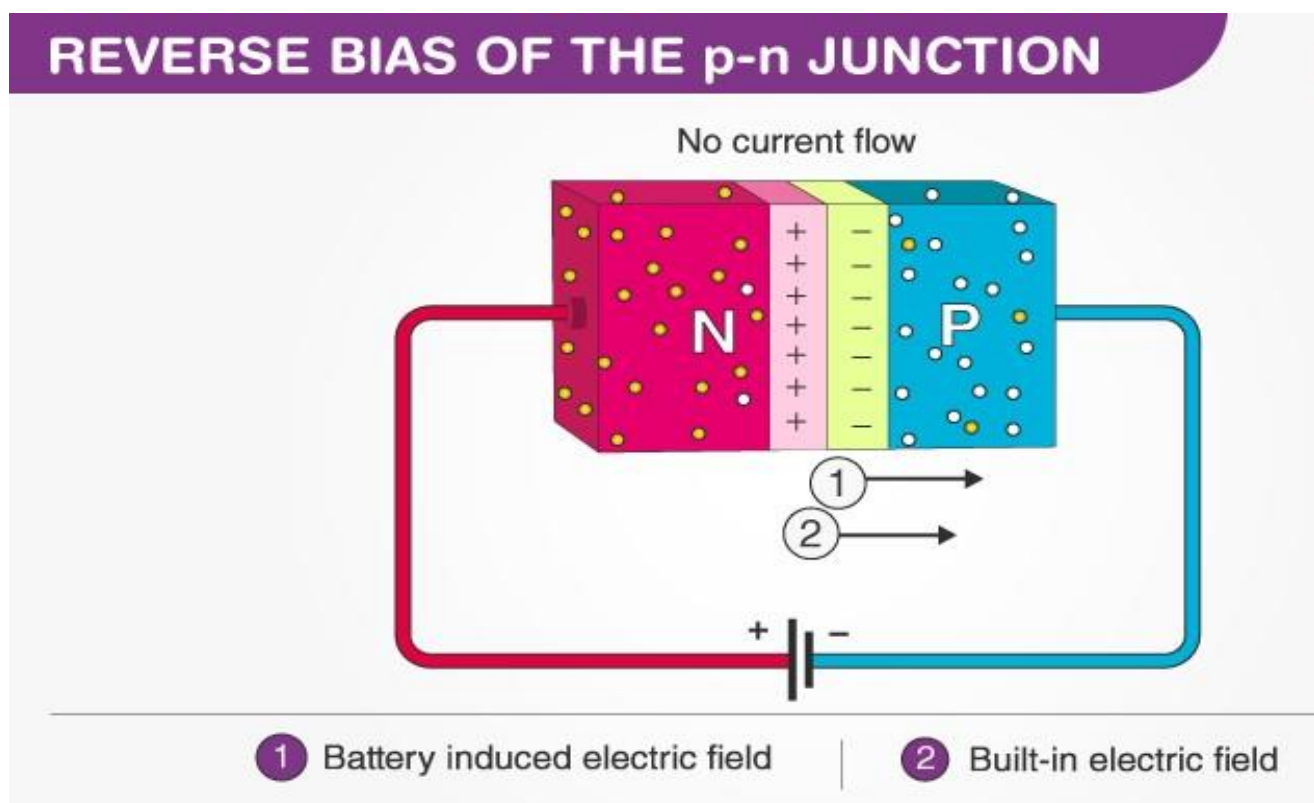
- Zero bias: No external voltage is applied to the P-N junction diode.
- Forward bias: The positive terminal of the voltage potential is connected to the p-type while the negative terminal is connected to the n-type.
- Reverse bias: The negative terminal of the voltage potential is connected to the p-type and the positive is connected to the n-type.

Forward Bias



When the p-type is connected to the battery's positive terminal and the n-type to the negative terminal, then the P-N junction is said to be forward-biased. When the P-N junction is forward biased, the built-in electric field at the P-N junction and the applied electric field are in opposite directions. When both the electric fields add up, the resultant electric field has a magnitude lesser than the built-in electric field. This results in a less resistive and thinner depletion region. The depletion region's resistance becomes negligible when the applied voltage is large. In silicon, at the voltage of 0.6 V, the resistance of the depletion region becomes completely negligible, and the current flows across it unimpeded.

Reverse Bias



When the p-type is connected to the battery's negative terminal and the n-type is connected to the positive side, the P-N junction is reverse biased. In this case, the built-in electric field and the applied electric field are in the same direction. When the two fields are added, the resultant electric field is in the same direction as the built-in electric field, creating a more resistive, thicker depletion region. The depletion region becomes more resistive and thicker if the applied voltage becomes larger.

P-N Junction Formula

The formula used in the P-N junction depends upon the built-in potential difference created by the electric field is given as:

$$E_0 = V_T \ln \left[\frac{N_D \cdot N_A}{n_i^2} \right]$$

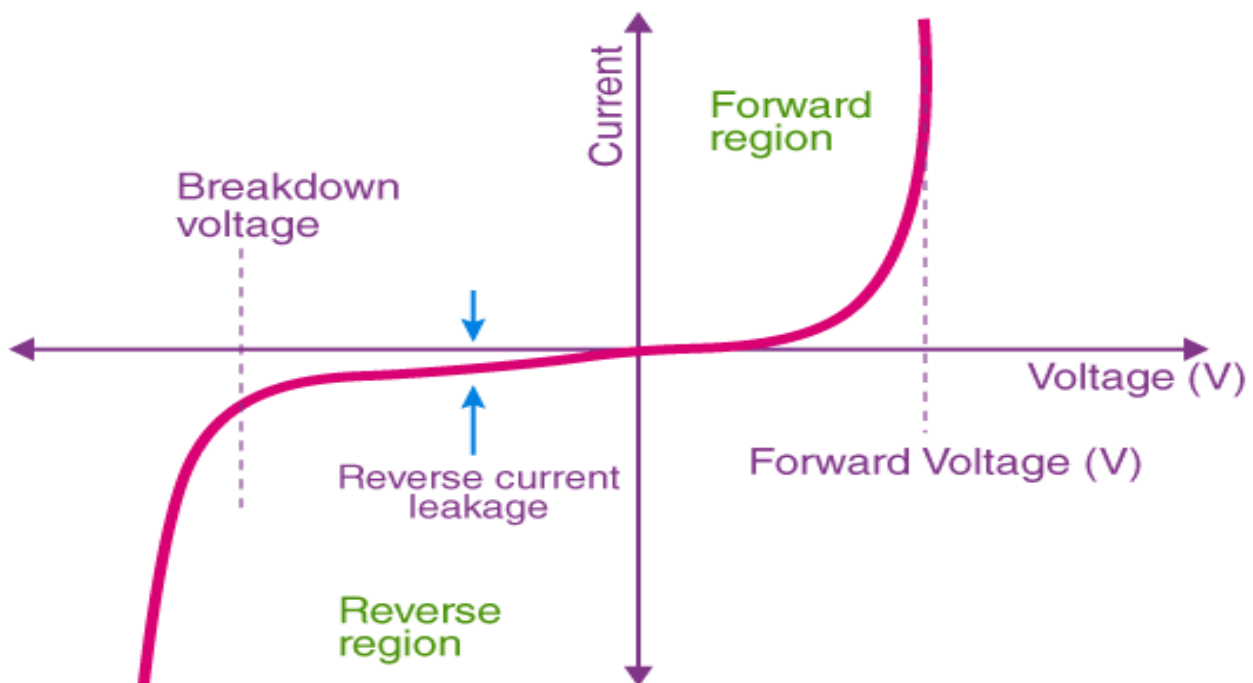
Where,

- E_0 is the zero bias junction voltage
- V_T is the thermal voltage of 26mV at room temperature
- N_D and N_A are the impurity concentrations
- n_i is the intrinsic concentration.

How does current flow in the PN junction diode?

The flow of electrons from the n-side towards the p-side of the junction takes place when there is an increase in the voltage. Similarly, the flow of holes from the p-side towards the n-side of the junction takes place along with the increase in the voltage. This results in the concentration gradient between both sides of the terminals. Due to the concentration gradient formation, charge carriers will flow from higher-concentration regions to lower-concentration regions. The movement of charge carriers inside the P-N junction is the reason behind the current flow in the circuit.

V-I Characteristics of P-N Junction Diode



VI characteristics of P-N junction diodes is a curve between the voltage and current through the circuit. Voltage is taken along the x-axis while the current is taken along the y-axis. The above graph is the V-I characteristics curve of the P-N junction diode. With the help of the curve, we can understand that there are three regions in which the diode works, and they are:

- Zero bias
- Forward bias
- Reverse bias

When the P-N junction diode is in zero bias condition, there is no external voltage applied and this means that the potential barrier at the junction does not allow the flow of current.

When the P-N junction diode is in forward bias condition, the p-type is connected to the positive terminal while the n-type is connected to the negative terminal of the external voltage. When the diode is arranged in this manner, there is a reduction in the potential barrier. For silicone diodes, when the voltage is 0.7 V and for germanium diodes, when the voltage is 0.3 V, the potential barriers decrease, and there is a flow of current.

When the diode is in forward bias, the current increases slowly, and the curve obtained is non-linear as the voltage applied to the diode overcomes the potential barrier. Once the diode overcomes the potential barrier, the diode behaves normally, and the curve rises sharply as the external voltage increases, and the curve obtained is linear.

When the P-N junction diode is in negative bias condition, the p-type is connected to the negative terminal while the n-type is connected to the positive terminal of the external voltage. This results in an increase in the potential barrier. Reverse saturation current flows in the beginning as minority carriers are present in the junction.

When the applied voltage is increased, the minority charges will have increased kinetic energy which affects the majority charges. This is the stage when the diode breaks down. This may also destroy the diode.

Applications of P-N Junction Diode

- P-N junction diode can be used as a photodiode as the diode is sensitive to the light when the configuration of the diode is reverse-biased.
- It can be used as a solar cell.
- When the diode is forward-biased, it can be used in LED lighting applications.
- It is used as rectifier in many electric circuits and as a voltage-controlled oscillator in varactors.

Physics Related Articles

Semiconductor Diode	Zener Diode
Extrinsic Semiconductors	Electric Field Lines

Frequently Asked Questions – FAQs

What happens when the battery voltage is increased in a forward-biased P-N junction?

The current through the junction increases when the battery voltage is increased in a forward-biased P-N junction.

What happens when a P-N junction is reverse biased?

The holes and electrons tend to move away from the junction.

What are the two breakdown mechanisms of the P-N junction?

The two breakdown mechanisms are Zener breakdown and Avalanche breakdown.

What is the static resistance of a diode?

Static resistance of a diode is defined as the ratio of the DC voltage applied across the diode to the DC current flowing through the diode.

What is the dynamic resistance of a diode?

Dynamic resistance of a diode is defined as the ratio of change in voltage to the change in current.

What is reverse resistance?

Reverse resistance is defined as the resistance offered by the P-N junction diode when it is reverse biased.

What is a semiconductor?

A semiconductor is a material whose conductivity stays between an insulator and a conductor.

What are the two types of semiconductors?

N-type semiconductors and p-type semiconductors are the two types of semiconductors.

What are n-type semiconductors?

N-type semiconductors are intrinsic semiconductors doped with antimony (Sb), phosphorus (P), or arsenic (As) as doping impurities.

What is meant by doping?

In semiconductor technology, doping is the process of intentional infliction of impurities into intrinsic semiconductors for the objective of remodelling their optical, structural, and electrical properties.

What is a diode?

A diode is an electrical device that enables the current to flow only in one direction. In an electrical circuit diagram, a diode is denoted by a triangle symbol with a line along one vertex.

What are the main types of diodes?

Light-emitting diode, avalanche diode, laser diode, Schottky diode, photodiode, p-n junction diode, and Zener diode are the main types of diodes.

What are the three regions in which diodes function?

Forward bias, zero bias and reverse bias are the three regions in which diodes function.

Give one application of P-N junction diodes.

P-N junction diodes are used as rectifiers in numerous electric circuits. They are also used as voltage-controlled oscillators in varactors.

What is Rectifier?

A rectifier is an electronic device that converts an alternating current into a direct current by using one or more P-N junction diodes. A diode behaves as a one-way valve that allows current to flow in a single direction. This process is known as rectification.

A rectifier can take the shape of several physical forms such as solid-state diodes, vacuum tube diodes, mercury-arc valves, silicon-controlled rectifiers, and various other silicon-based semiconductors switches.

Different Types of Rectifier

Rectifiers are mainly classified into two types as:

1. Uncontrolled Rectifier
2. Controlled Rectifier

Uncontrolled Rectifiers

The type of rectifier whose voltage cannot be controlled is known as an uncontrolled rectifier. Uncontrolled rectifiers are further divided as follows:

- Half Wave Rectifier
- Full Wave Rectifier

The type of rectifier that converts only the half cycle of the alternating current into the direct current is known as a half-wave rectifier. Likewise, a full-wave rectifier converts both positive and negative half cycles of the AC. An example of this is a bridge rectifier. A bridge rectifier uses 4 diodes that are connected in the form of a Wheatstone bridge.

Controlled Rectifiers

A type of rectifier whose voltage can be varied is known as the controlled rectifier. We use SCRs, MOSFETs and IGBTs to make an uncontrolled rectifier a controlled one. These rectifiers are preferred over their uncontrolled counterparts. There are two types of controlled rectifiers, and they are Half Wave Controlled Rectifier and Full Wave Controlled Rectifier. Half-wave controlled rectifier has the same design as the half-wave uncontrolled rectifier except we replace the diode with an SCR.

- Full Wave Rectifier

- Half Wave Rectifier

What Are Some Applications of Rectifiers?

Some common applications of rectifiers are:

- Rectifiers are used in electric welding to provide polarized voltage
- Half-wave rectifiers are used as a mosquito repellent
- Half-wave rectifiers are used as a signal peak detector in AM radio
- Rectifiers are used in modulation, demodulation and voltage multipliers

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Frequently Asked Questions – FAQs

What is a rectifier?

A rectifier is an electronic device that converts an alternating current into a direct current by using one or more P-N junction diodes.

What are the types of rectifiers?

The following are the types of rectifiers:

- Uncontrolled Rectifier
- Controlled Rectifier

What is a half-wave rectifier?

The type of rectifier that converts only the half cycle of the alternating current into the direct current is known as a half-wave rectifier.

State true or false: Half-wave rectifiers are used as a mosquito repellent.

True.

List three applications of rectifiers.

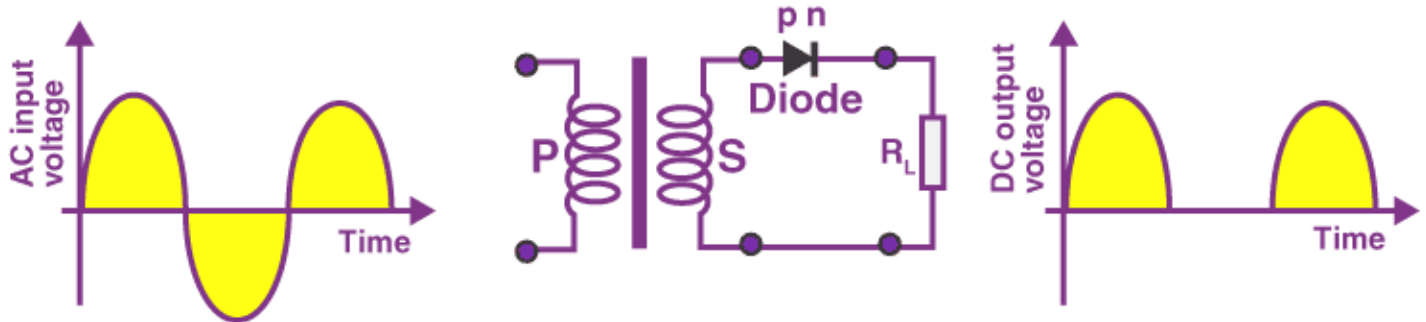
- ☐ Rectifiers are used in modulation, demodulation and voltage multipliers.
- ☐ Half-wave rectifiers are used as a signal peak detector in AM radio.
- ☐ Rectifiers are used in electric welding to provide polarized voltage

What is Half Wave Rectifier?

In a half-wave rectifier, one half of each a.c input cycle is rectified. When the p-n junction diode is forward biased, it gives little resistance and when it is reversed biased it provides high resistance. During one-half cycles, the diode is forward biased when the input voltage is applied and in the opposite half cycle, it is reverse biased. During alternate half-cycles, the optimum result can be obtained.

Working of Half Wave Rectifier

The half-wave rectifier has both positive and negative cycles. During the positive half of the input, the current will flow from positive to negative which will generate only a positive half cycle of the a.c supply. When a.c supply is applied to the transformer, the voltage will be decreasing at the secondary winding of the diode. All the variations in the a.c supply will reduce, and we will get the pulsating d.c voltage to the load resistor.



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In the second half cycle, the current will flow from negative to positive and the diode will be reverse biased. Thus, at the output side, there will be no current generated, and we cannot get power at the load resistance. A small amount of reverse current will flow during reverse bias due to minority carriers.

Characteristics of Half Wave Rectifier

Following are the characteristics of half-wave rectifier:

Ripple Factor

Ripples are the oscillations that are obtained in DC which are corrected by using filters such as inductors and capacitors. These ripples are measured with the help of the ripple factor and are denoted by γ . Ripple factor tells us the number of ripples presents in the output DC. Higher the ripple factor, more is the oscillation at the output DC and lower is the ripple factor, less is the oscillation at the output DC.

Ripple factor is the ratio of RMS value of the AC component of the output voltage to the DC component of the output voltage.

$$\gamma = \sqrt{V_{rms}^2 - V_{DC}^2} / V_{DC}$$

DC Current

DC current is given as:

$$I_{DC} = I_{max} / \pi$$

Where,

- I_{max} is the maximum DC load current

DC Output Voltage

The output DC voltage appears at the load resistor R_L which is obtained by multiplying output DC voltage with the load resistor R_L . The output DC voltage is given as:

$$V_{DC} = V_{Smax} / \pi$$

Where,

- V_{Smax} is the maximum secondary voltage

Form Factor

The form factor is the ratio of RMS value to the DC value. **For a half-wave rectifier, the form factor is 1.57.**

Rectifier Efficiency

Rectifier efficiency is the ratio of output DC power to the input AC power. **For a half-wave rectifier, rectifier efficiency is 40.6%.**

Advantages of Half Wave Rectifier

- Affordable
- Simple connections
- Easy to use as the connections are simple
- Number of components used are less

Disadvantages of Half Wave Rectifier

- Ripple production is more
- Harmonics are generated
- Utilization of the transformer is very low
- The efficiency of rectification is low

Applications of Half Wave Rectifier

Following are the uses of half-wave rectification:

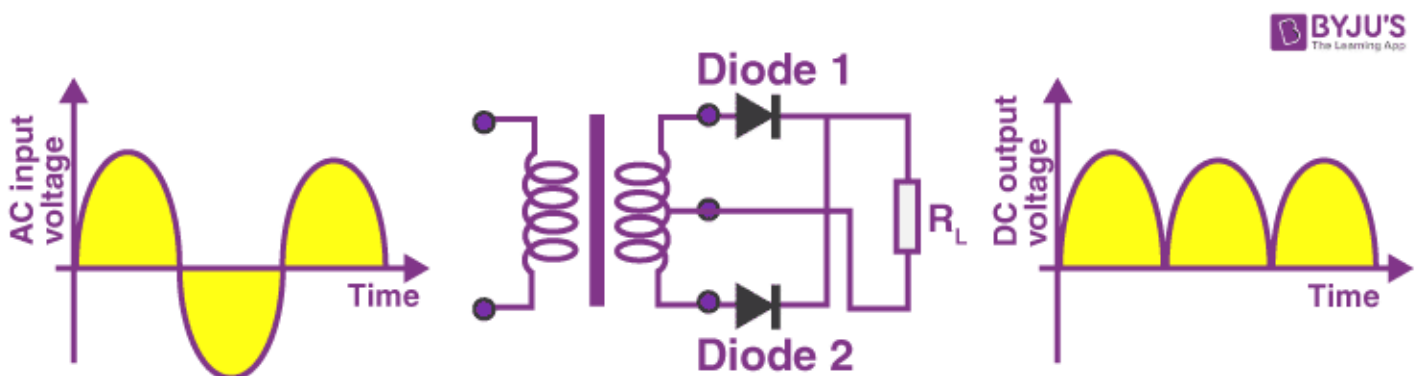
- **Power rectification:** Half wave rectifier is used along with a transformer for power rectification as powering equipment.
- **Signal demodulation:** Half wave rectifiers are used for demodulating the AM signals.
- **Signal peak detector:** Half wave rectifier is used for detecting the peak of the incoming waveform.

What is Full Wave Rectifier?

Full-wave rectifier circuits are used for producing an output voltage or output current which is purely DC. The main advantage of a full-wave rectifier over half-wave rectifier is that such as the average output voltage is higher in full-wave rectifier, there is less ripple produced in full-wave rectifier when compared to the half-wave rectifier.

Working of Full Wave Rectifier

The full-wave rectifier utilizes both halves of each a.c input. When the p-n junction is forward biased, the diode offers low resistance and when it is reverse biased it gives high resistance. The circuit is designed in such a manner that in the first half cycle if the diode is forward biased then in the second half cycle it is reverse biased and so on.



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Characteristics of Full Wave Rectifier

Following are the characteristics of full-wave rectifier:

Ripple Factor

Ripple factor for a full-wave rectifier is given as:

$$\gamma = (\frac{V_{rms}}{V_{DC}})^2 - 1$$

DC Current

Currents from both the diodes D_1 and D_2 are in the same direction when they flow towards load resistor R_L . The current produced by both the diodes is the ratio of I_{max} to π , therefore the DC current is given as:

$$I_{DC} = \frac{2I_{max}}{\pi}$$

Where,

- I_{max} is the maximum DC load current

DC Output Voltage

DC output voltage is obtained at the load resistor R_L and is given as:

$$V_{DC} = \frac{2V_{max}}{\pi}$$

Where,

- V_{max} is the maximum secondary voltage

Form Factor

The form factor is the ratio of RMS value of current to the output DC voltage. The form ***factor of a full-wave rectifier is given as 1.11***

Rectifier Efficiency

Rectifier efficiency is used as a parameter to determine the efficiency of the rectifier to convert AC into DC. It is the ratio of DC output power to the AC input power. The rectifier ***efficiency of a full-wave rectifier is 81.2%***.

Types of Full Wave Rectifier

There are two main types of full-wave rectifiers, and they are:

- Two diodes full-wave rectifier circuit** (requires a center-tapped transformer and is used in vacuum tubes)
- Bridge rectifier circuit** (doesn't require a centre-tapped transformer and is used along with transformers for efficient usage)

Advantages of Full Wave Rectifier

- The rectifier efficiency of a full-wave rectifier is high
- The power loss is very low
- Number of ripples generated are less

Disadvantages of Full Wave Rectifier

- Very expensive

Applications of Full Wave Rectifier

Following are the uses of full-wave rectifier:

- Full-wave rectifiers are used for supplying polarized voltage in welding and for this bridge rectifiers are used.
- Full-wave rectifiers are used for detecting the amplitude of modulated radio signals.

Difference between Half Wave Rectifier and Full Wave Rectifier

Parameter	Half Wave Rectifier	Full Wave Rectifier
Definition	The half-wave rectifier is a rectifier which is used for converting the one-half cycle of AC input to DC output	A full-wave rectifier is a rectifier which is used for converting both the half cycles of AC input into DC output
No. of diodes used	1	2 or 4 depending on the type of circuit
Form factor	1.57	1.11
Rectifier efficiency	40.6%	81.2%
Ripple factor	Ripple factor of a half-wave rectifier is more	Ripple factor of a full-wave rectifier is less

Frequently Asked Questions – FAQs

Why half-wave rectifiers are not used in dc power supply?

Half-wave rectifiers are not used in dc power supply because the supply provided by the half-wave rectifier is not satisfactory.

What is PIV of a diode in a rectifier circuit?

PIV stands for peak inverse voltage and it is the maximum voltage that is possible to occur across the diode when it is operated in reverse biased.

What is a ripple in a rectifier circuit?

Ripple is defined as the ac component that has a pulsating output in a rectifier.

What are the advantages of a bridge rectifier over a center-tapped full-wave rectifier?

Following are the three advantages of bridge rectifier over a center-tapped full-wave rectifier:

- The TUF of a bridge rectifier is 81.2% while the TUF of a center-tapped is 67.2%.
- The output of the bridge rectifier is twice that of the center-tapped full-wave rectifier.
- A bridge rectifier as a PIV half of the center-tapped full-wave rectifier.

What is the cycle followed by the current in a full-wave rectifier?

In a full-wave rectifier, the current flows in the half cycle of the input signal.

What happens to the peak current if the value of the shunt capacitor filter is increased?

When the value of the shunt capacitor filter is increased, the peak current will also increase in a rectifying diode.

Why bridge rectifier is preferred over an ordinary two diodes full-wave rectifier?

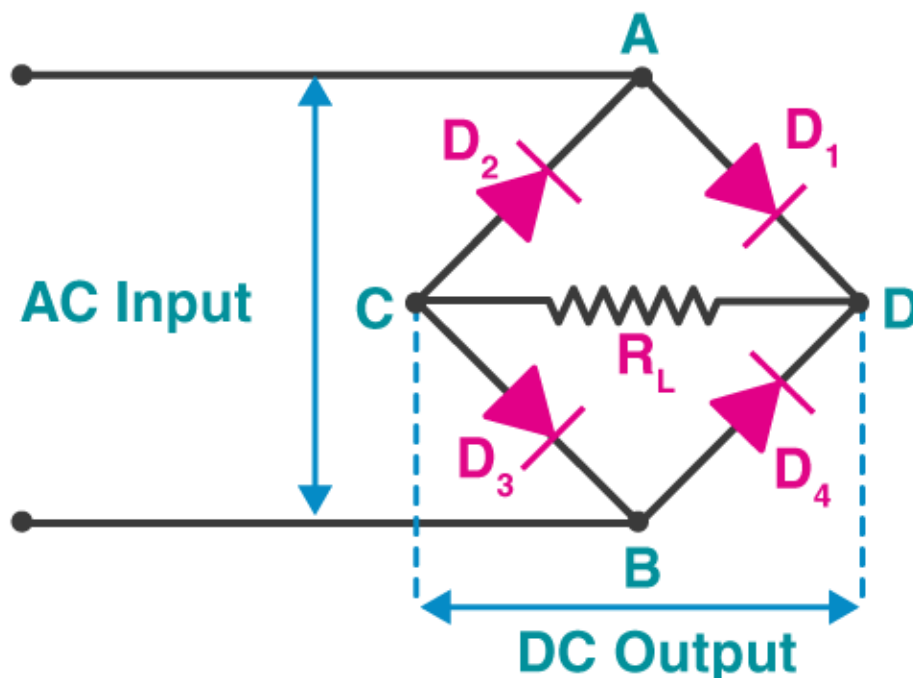
A bridge rectifier is preferred over an ordinary two diodes full-wave rectifier because:

- The PIV is less per diode
- There is no need for the centre tap
- The transformer required is smaller with the same output

BRIDGE RECTIFIER

Construction

The construction of a bridge rectifier is shown in the figure below. The bridge rectifier circuit is made of four diodes D_1 , D_2 , D_3 , D_4 and a load resistor R_L . The four diodes are connected in a closed-loop configuration to efficiently convert the alternating current (AC) into Direct Current (DC). The main advantage of this configuration is the absence of the expensive centre-tapped transformer. Therefore, the size and cost are reduced.

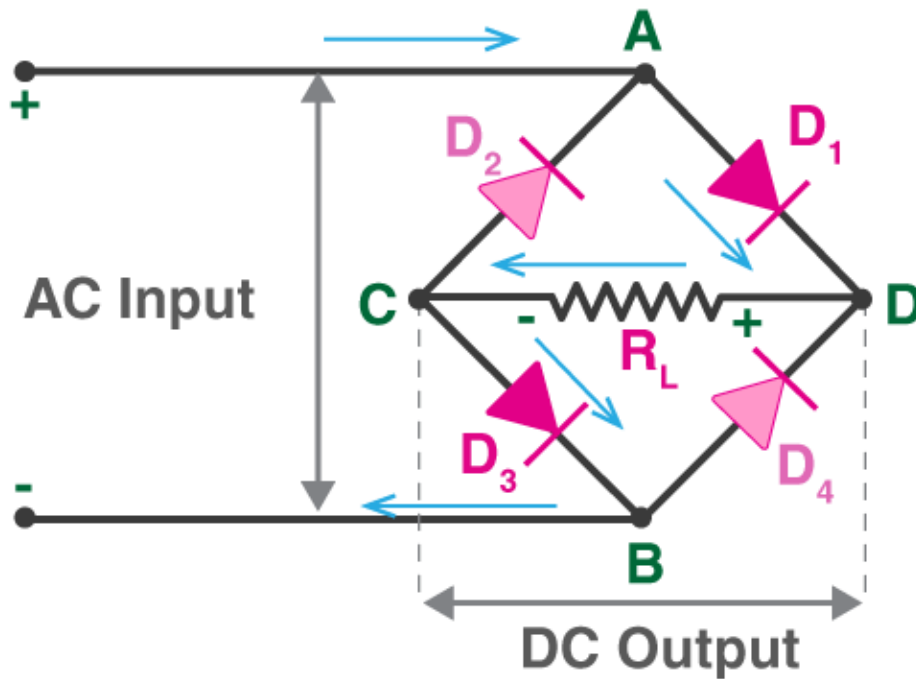


The input signal is applied across terminals A and B, and the output DC signal is obtained across the load resistor R_L connected between terminals C and D. The four diodes are arranged in such a way that only two diodes conduct electricity during each half cycle. D_1 and D_3 are pairs that conduct electric current during the positive half cycle/. Likewise, diodes D_2 and D_4 conduct electric current during a negative half cycle.

Working

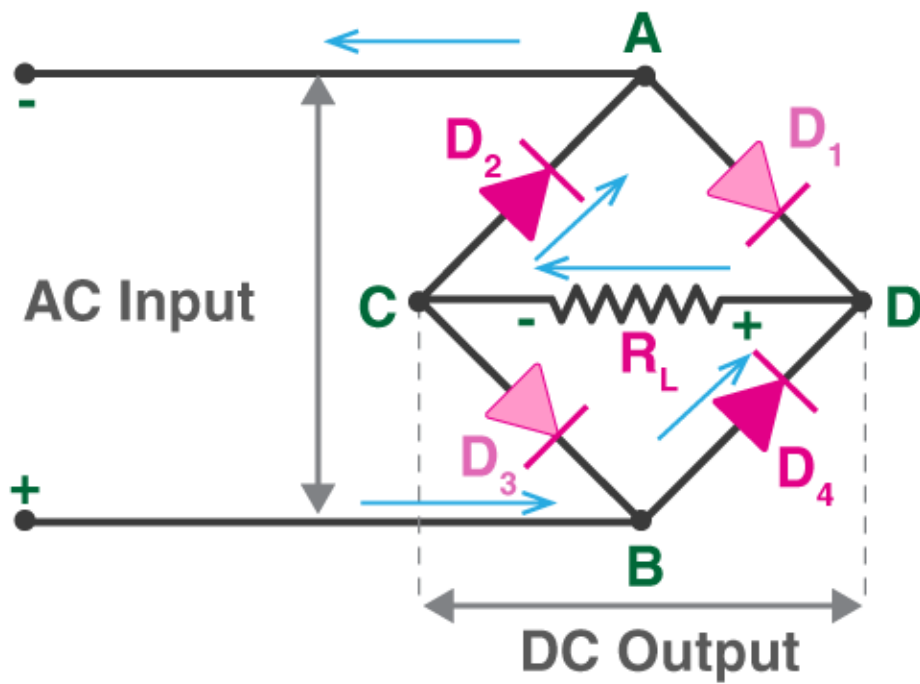
When an AC signal is applied across the bridge rectifier, terminal A becomes positive during the positive half cycle while terminal B becomes negative. This results in diodes D_1 and D_3 becoming forward biased while D_2 and D_4 becoming reverse biased.

The current flow during the positive half-cycle is shown in the figure below:



During the negative half-cycle, terminal B becomes positive while terminal A becomes negative. This causes diodes D_2 and D_4 to become forward biased and diode D_1 and D_3 to be reverse biased.

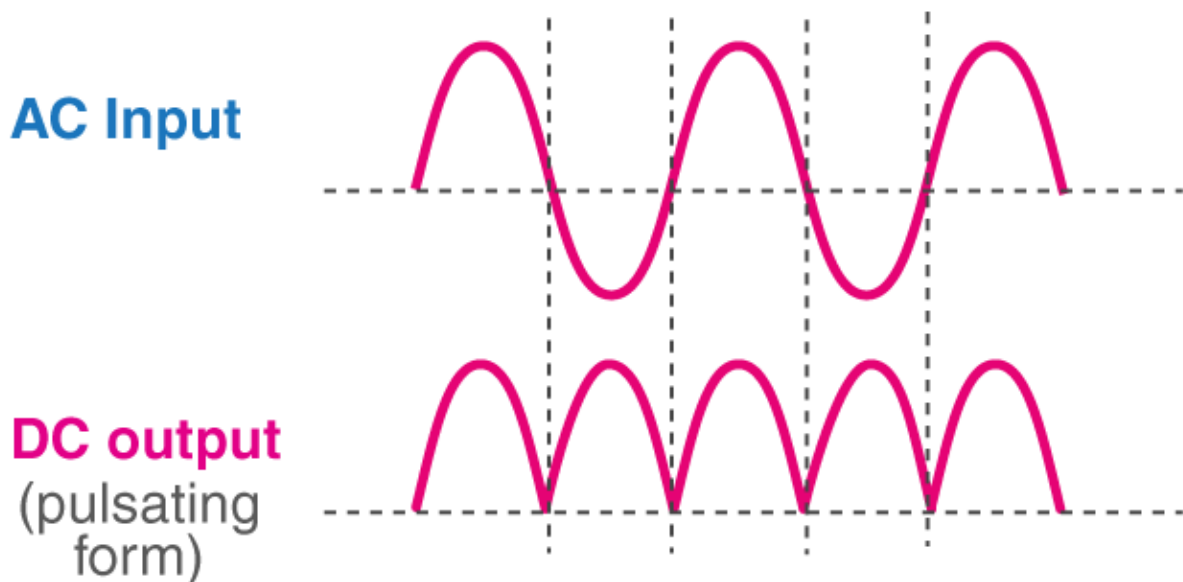
The current flow during the negative half cycle is shown in the figure below:



From the figures given above, we notice that the current flow across load resistor R_L is the same during the positive and negative half-cycles. The output DC signal polarity may be either completely positive or negative. In our case, it is completely positive. If the diodes' direction is reversed, we get a complete negative DC voltage.

Thus, a bridge rectifier allows electric current during both positive and negative half cycles of the input AC signal.

The output waveforms of the bridge rectifier are shown in the below figure.



Characteristics of Bridge Rectifier

Ripple Factor

The smoothness of the output DC signal is measured by a factor known as the ripple factor. The output DC signal with fewer ripples is considered a smooth DC signal while the output with high ripples is considered a high pulsating DC signal.

Mathematically, the ripple factor is defined as the ratio of ripple voltage to pure DC voltage.

The ripple factor for a bridge rectifier is given by

$$\gamma = \frac{V_{rms}}{V_{DC}} - 1$$

For bridge rectifiers, the ripple factor is 0.48.

Peak Inverse Voltage

The maximum voltage that a diode can withstand in the reverse bias condition is known as a peak inverse voltage. During the positive half cycle, the diodes D_1 and D_3 are in the conducting state while D_2 and D_4 are in the non-conducting state. Similarly, during the negative half cycle, diodes D_2 and D_4 are in the conducting state, and diodes D_1 and D_3 are in the non-conducting state.

Efficiency

The rectifier efficiency determines how efficiently the rectifier converts Alternating Current (AC) into Direct Current (DC). Rectifier efficiency is defined as the ratio of the DC output power to the AC input power. The maximum efficiency of a bridge rectifier is 81.2%.

$$\eta = \frac{P_{DC}}{P_{AC}}$$

Advantages

- The efficiency of the bridge rectifier is higher than the efficiency of a half-wave rectifier. However, the rectifier efficiency of the bridge rectifier and the centre-tapped full-wave rectifier is the same.
- The DC output signal of the bridge rectifier is smoother than the output DC signal of a half-wave rectifier.
- In a half-wave rectifier, only half of the input AC signal is used, and the other half is blocked. Half of the input signal is wasted in a half-wave rectifier. However, in a bridge rectifier, the electric current is allowed during both positive and negative half cycles of the input AC signal. Hence, the output DC signal is almost equal to the input AC signal.

Disadvantages

- The circuit of a bridge rectifier is complex when compared to a half-wave rectifier and centre-tapped full-wave rectifier. Bridge rectifiers use 4 diodes while half-wave rectifiers and centre-tapped full wave rectifiers use only two diodes.
- When more diodes are used more power loss occurs. In a centre-tapped full-wave rectifier, only one diode conducts during each half cycle. But in a bridge rectifier, two diodes connected in series conduct during each half cycle. Hence, the voltage drop is higher in a bridge rectifier.

Frequently Asked Questions – FAQs

Choose YES or No: Can rectifiers convert an AC power to a DC power?

YES.

Define bridge rectifiers.

The bridge rectifier is a type of full-wave rectifier that uses four or more diodes in a bridge circuit configuration to convert alternating (AC) current to a direct (DC) current.

What is a rectifier?

A rectifier is an electronic device that converts an alternating current into a direct current by using one or more P-N junction diodes.

State true or false: The bridge rectifier allows electric current flow during both negative and positive half cycles of the input AC signal.

TRUE.

What is the maximum efficiency of a bridge rectifier?

The maximum efficiency of a bridge rectifier is 81.2%.

Zener Diode

A Zener diode not only allows current to flow from anode to cathode but also, in the reverse direction on reaching the Zener voltage. Due to this functionality, Zener diodes are the most commonly used semiconductor diodes. In this article, let us learn the function of Zener diodes along with its construction, operation and more.

Zener Diode Explanation

A Zener Diode, also known as a breakdown diode, is a heavily doped semiconductor device that is designed to operate in the reverse direction. When the voltage across the terminals of a Zener diode is reversed and the potential reaches the Zener Voltage (knee voltage), the junction breaks down and the current flows in the reverse direction. This effect is known as the Zener Effect.

Zener Diode Definition

A Zener diode is a heavily doped semiconductor device that is designed to operate in the reverse direction.

Zener diodes are manufactured with a great variety of Zener voltages (V_z) and some are even made variable.

How does a Zener Diode work in reverse bias?

A Zener diode operates just like a normal diode when it is forward-biased. However, a small leakage current flows through the diode when connected in reverse biased mode. As the reverse voltage increases to the predetermined breakdown voltage (V_z), current starts flowing through the diode. The current increases to a maximum, which is determined by the series resistor, after which it stabilizes and remains constant over a wide range of applied voltage.

There are two types of breakdowns for a Zener Diode:

- - Avalanche Breakdown
 - Zener Breakdown

Avalanche Breakdown in Zener Diode

Avalanche breakdown occurs in normal diode and Zener Diode at high reverse voltage. When a high value of reverse voltage is applied to the PN junction, the free electrons gain sufficient energy and accelerate at high velocities. These free electrons moving at high velocity collide with other atoms and knock off more electrons. Due to this continuous collision, a large number of free electrons are generated as a result of electric current in the diode rapidly increases. This sudden increase in electric current may permanently destroy the normal diode.

However, a Zener diode is designed to operate under avalanche breakdown and can sustain the sudden spike of current. Avalanche breakdown occurs in Zener diodes with Zener voltage (V_z) greater than 6V.

Zener Breakdown in Zener Diode

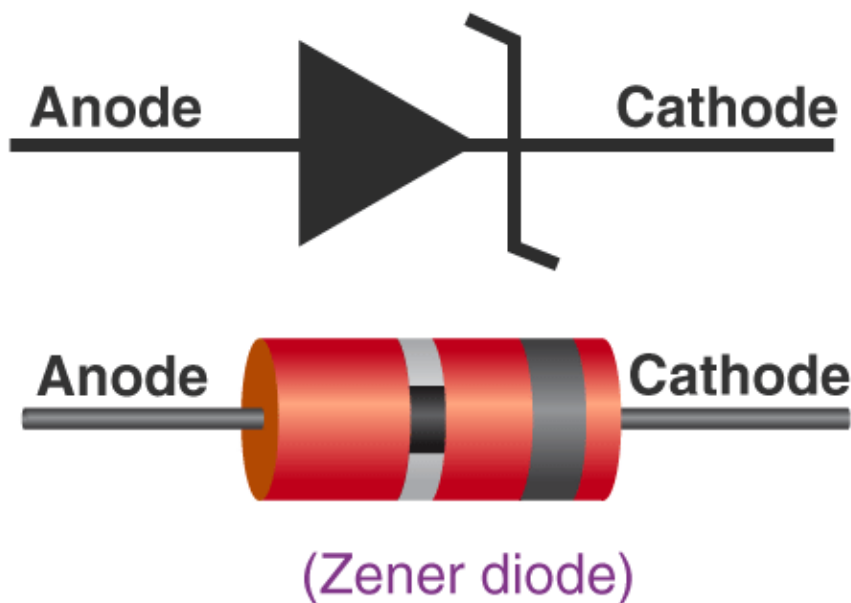
When the applied reverse bias voltage reaches closer to the Zener voltage, the electric field in the depletion region gets strong enough to pull electrons from their valence band. The valence electrons that gain sufficient energy from the strong electric field of the depletion region break free from the parent atom. At the Zener breakdown region, a small increase in the voltage results in the rapid increase of the electric current.

Avalanche Breakdown vs Zener Breakdown

- The Zener effect is dominant in voltages up to 5.6 volts and the avalanche effect takes over above that.
- They are both similar effects, the difference being that the Zener effect is a quantum phenomenon and the avalanche effect is the movement of electrons in the valence band like in any electric current.
- Avalanche effect also allows a larger current through the diode than what a Zener breakdown would allow.

Circuit Symbol of Zener Diode

There are many ways in which a Zener diode is packaged. Some are used for high levels of power dissipation and the others are contained with surface mount formats. The most common Zener diode type is contained within a small glass encapsulation. It has a band around one end marking the cathode side of the diode.



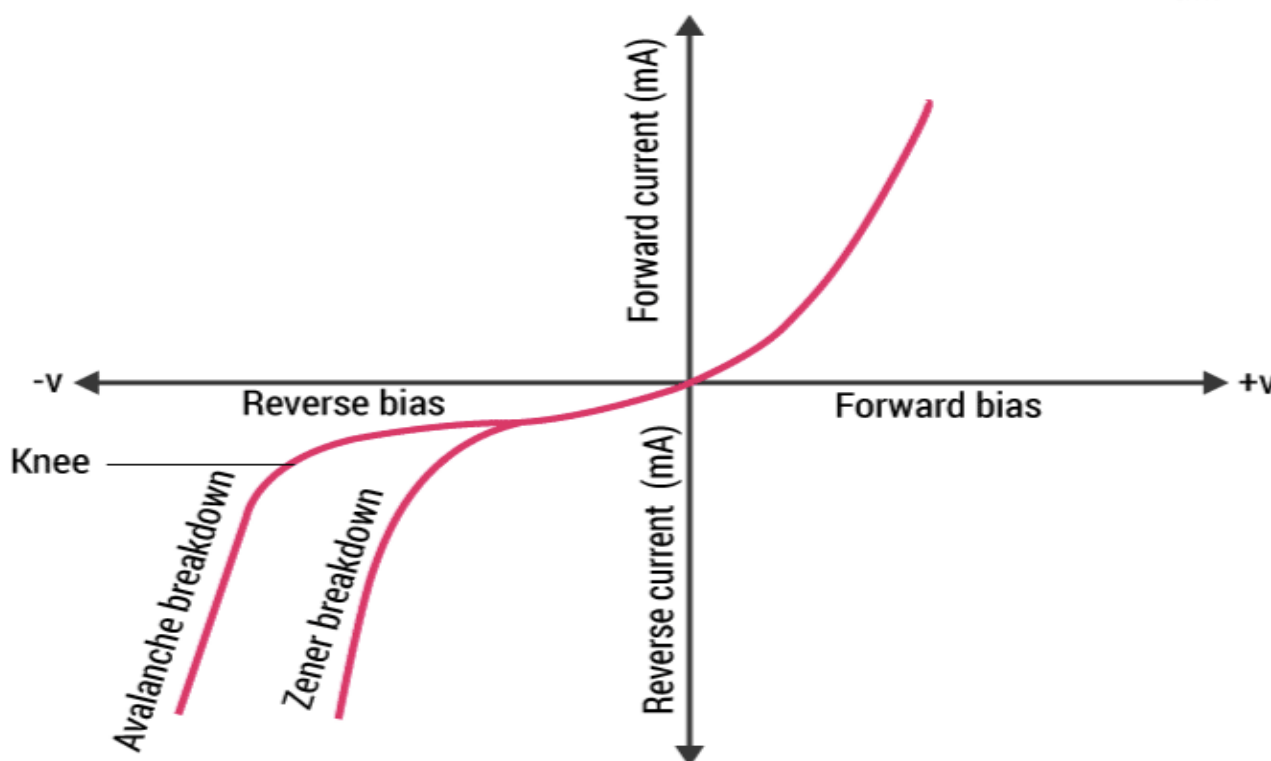
Zener diode symbol and package outlines

From the diagram, we can see that the band around the package corresponds to the line on the diode circuit symbol and this can be an easy way of remembering which end is for which.

The Zener diode circuit symbol places two tags at the end of the bar – one in the upward direction and the other in the lower direction, as shown in the figure. This helps in distinguishing Zener diodes from other forms of diodes within the circuit.

V-I Characteristics of Zener Diode

The diagram given below shows the V-I characteristics of the Zener diode.



When reverse-biased voltage is applied to a Zener diode, it allows only a small amount of leakage current until the voltage is less than Zener voltage.

The V-I characteristics of a Zener diode can be divided into two parts as follows:

- (i) Forward Characteristics
- (ii) Reverse Characteristics

Forward Characteristics of Zener Diode

The first quadrant in the graph represents the forward characteristics of a Zener diode. From the graph, we understand that it is almost identical to the forward characteristics of any other P-N junction diode.

Reverse Characteristics of Zener Diode

When a reverse voltage is applied to a Zener voltage, a small reverse saturation current I_o flows across the diode. This current is due to thermally generated minority carriers. As the reverse voltage increases, at a certain value of reverse voltage, the reverse current increases drastically and sharply. This is an indication that the breakdown has occurred. We call this voltage breakdown voltage or Zener voltage, and V_z denotes it.

Zener Diode Specifications

Some commonly used specifications for Zener diodes are as follows:

- **Zener/Breakdown Voltage** – The Zener or the reverse breakdown voltage ranges from 2.4 V to 200 V, sometimes it can go up to 1 kV while the maximum for the surface-mounted device is 47 V.
- **Current I_z (max)** – It is the maximum current at the rated Zener Voltage ($V_z - 200\mu\text{A}$ to 200 A)
- **Current I_z (min)** – It is the minimum value of current required for the diode to breakdown.
- **Power Rating** – It denotes the maximum power the Zener diode can dissipate. It is given by the product of the voltage of the diode and the current flowing through it.
- **Temperature Stability** – Diodes around 5 V have the best stability
- **Voltage Tolerance** – It is typically $\pm 5\%$
- **Zener Resistance (R_z)** – It is the resistance to the Zener diode exhibits.

Application of Zener Diode

Following are the applications of Zener diode:

Zener diode as a voltage regulator:

Zener diode is used as a Shunt voltage regulator for regulating voltage across small loads. The Zener diode is connected parallel to the load to make it reverse bias, and once the Zener diode exceeds knee voltage, the voltage across the load will become constant. The breakdown voltage of Zener diodes will be constant for a wide range of currents.

Zener diode in over-voltage protection:

When the input voltage is higher than the Zener breakage voltage, the voltage across the resistor drops resulting in a short circuit, this can be avoided by using the Zener diode.

Zener diode in clipping circuits:

Zener diode is used for modifying AC waveform clipping circuits by limiting the parts of either one or both the half cycles of an AC waveform.

Frequently Asked Questions – FAQs

How do you define Zener diode?

A Zener diode is a semiconductor device that allows current to flow either in a forward or reverse direction.

Why is Zener Diode used a regulator?

Zener diode is used as a shunt voltage regulator. Zener diode is connected parallel to the load to make it reverse bias and once the Zener diode exceeds knee voltage, the voltage across the load becomes constant.

Does Zener Diode exhibit a controlled breakdown?

Yes, Zener Diode does exhibit a controlled breakdown.

Does Zener Diode work in forward biased mode?

Yes, Zener Diode does exhibit a controlled breakdown.

What is the difference between a Zener diode and a normal diode?

The main difference between a Zener diode and a normal diode lies in the passage of current. A normal diode allows current to flow only in one direction while Zener diode allows current to flow in both directions.

What is a Zener diode also called as?

A Zener diode is also called as a breakdown diode.

Who was the first person to describe the electrical properties of Zener diode?

Clarence Melvin Zener was the first person to describe the electrical properties of Zener diode.

What are the types of breakdowns for a Zener Diode?

There are two types of breakdowns for a Zener Diode:

- Avalanche Breakdown
- Zener Breakdown

What is the voltage tolerance of a Zener diode?

Voltage tolerance of a Zener diode is typically $\pm 5\%$.

Uses Of Zener Diode

We are aware of the uses of the zener diode which is to permit the current flow in forward or reverse direction. Zener diode is a semiconductor device consisting of a P-N junction which conducts the current in reverse direction when a certain specified voltage is reached. It has a reverse breakdown voltage which means it can continuously operate in a reverse-bias mode without getting damaged.

Applications and Uses of Zener Diode can be seen in the following:

- As a voltage regulator
- Protects from overvoltage
- Used in clipping circuits
- Used to shift voltage

Since the voltage drop across the diode remains constant over a wide range of voltages, the zener diode is suitable for voltage regulation.

Zener diode acts as a voltage regulator

When the load voltage equals the breakdown voltage, the resistors which are connected in series limit the current through the diode when there is an excess amount of voltage while the diode is conducting. While doing that the diode generates some kind of noise which can be suppressed by adding a large value decoupling capacitor across the diode.

Read more about the zener diode as a voltage regulator from the link below

- [Zener diode as a voltage regulator](#)

Zener diode protects from over voltage

When the input voltage increases and reaches the Zener breakdown voltage, the current through the diode will create a voltage drop across the resistor. This triggers a short circuit to the ground.

Zener diode is used in clipping circuits

Clipper circuits are used to prevent the output signal from going beyond a determined value without changing anything of the input signal. Uses of zener diode include modifying and shaping the AC waveform clipping circuits. The clipping circuits limit one part of the AC waveform to shape the waveform or provide protection. These kinds of circuits are generally used in TV and FM transmitters to remove interference.

Zener diode can be used to shift voltage

Zener diode's core uses include converting a signal from one voltage to another. It has the ability to maintain a steady output voltage in the breakdown region. This voltage shifting functionality makes zener diode an ideal component of the operation. These were some uses of zener diode. To learn the uses of other electrical components, click on the link below

- [Uses of Diode](#)
- [Uses of Transistor](#)

What is a Voltage Regulator?

Appearance



Schematic Symbol



Backwards current flow too, but only past the “**Zener**” breakdown voltage

A Semiconductor Diode blocks current in the reverse direction but will suffer damage if the reverse voltage applied across becomes too high.

A voltage regulator is a device that regulates the voltage level. It essentially steps down the input voltage to the desired level and keeps it at that same level during the supply. This ensures that even when a load is applied the voltage doesn't drop. The voltage regulator is used for two main reasons, and they are:

- To vary or regulate the output voltage
- To keep the output voltage constant at the desired value in spite of variations in the supply voltage.

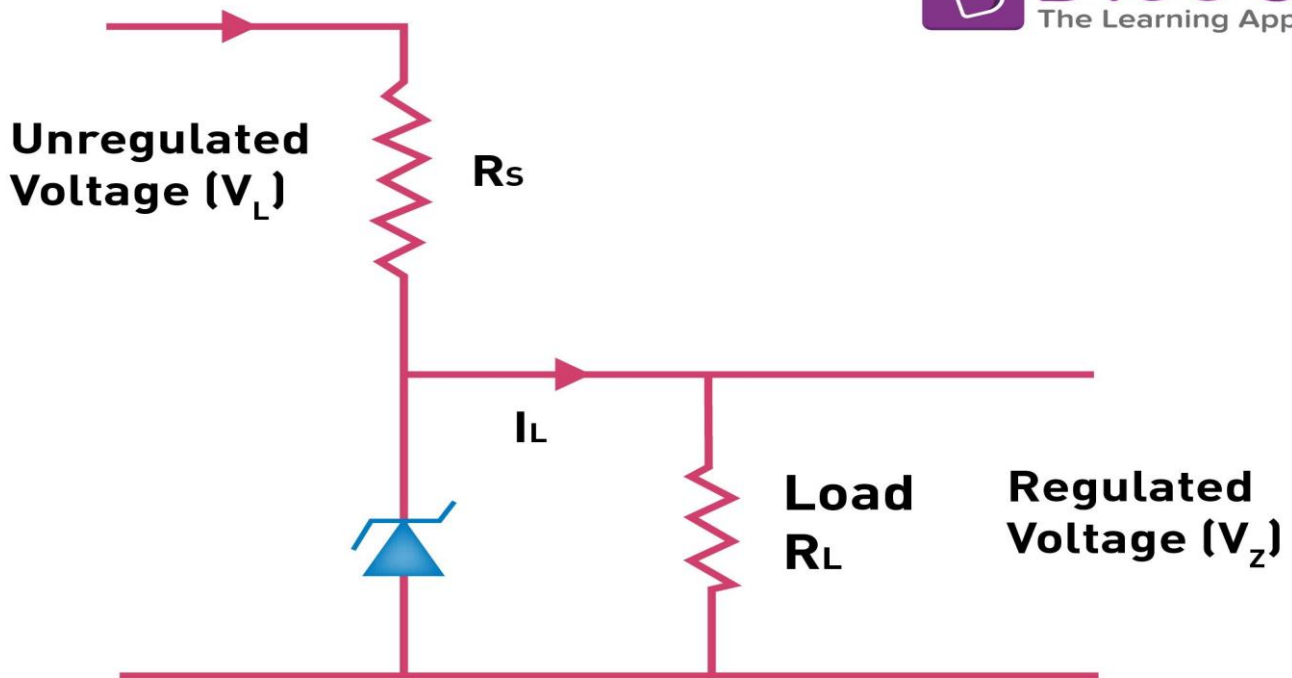
Voltage regulators are used in computers, power generators, alternators to control the output of the plant.

Zener Diode as a Voltage Regulator

There is a series resistor connected to the circuit in order to limit the current into the diode. It is connected to the positive terminal of the d.c. It works in such a way the reverse-biased can also work in breakdown conditions. We do not use ordinary junction diode because the low power rating diode can get damaged when we apply reverse bias above its breakdown voltage. When the minimum input voltage and the maximum load current is applied, the Zener diode current should always be minimum.

Since the input voltage and the required output voltage is known, it is easier to choose a Zener diode with a voltage approximately equal to the load voltage, i.e. $V_Z = V_L$.

The circuit diagram of a voltage regulator using a Zener diode is shown:



The value of the series resistor is written as $R_s = (V_L - V_z)I_L$.

Current through the diode increases when the voltage across the diode tends to increase which results in the voltage drop across the resistor. Similarly, the current through the diode decreases when the voltage across the diode tends to decrease. Here, the voltage drop across the resistor is very less, and the output voltage results normally.

Frequently Asked Questions – FAQs

Name the factor which is required for voltage regulation to keep diode in the breakdown region.

Minimum reverse current is required for voltage regulation to keep the diode in the breakdown region.

How to set the breakdown region during manufacturing of the diode?

By controlling the doping level, the breakdown region can be set during the manufacturing of the diode.

What is a Zener breakdown?

When the breakdown occurs in Rectifier as well as the Zener diode, it is known as the Zener breakdown.

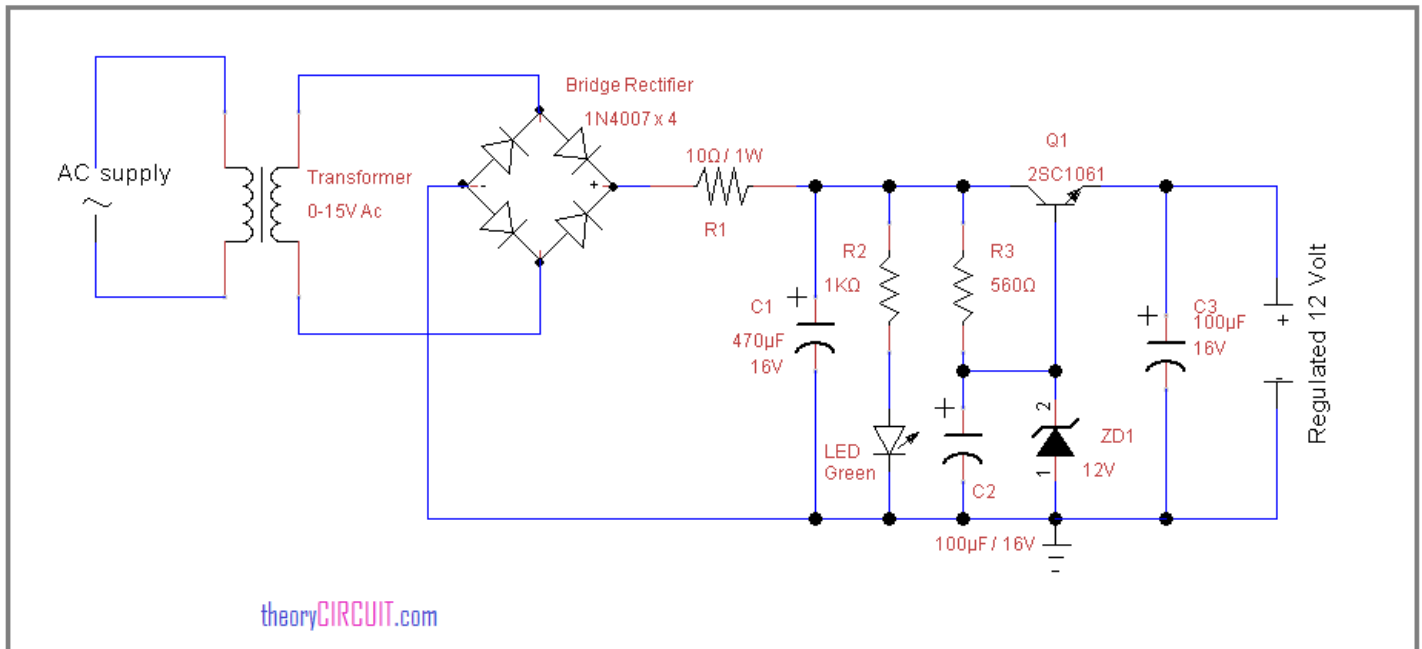
What happens to the DC output voltage when the input varies within the limits?

The DC output voltage remains constant when the input varies within the limits as the voltage regulator is used.

12 Volt Regulated power supply circuit using zener diode

Here this circuit constructed with bridge rectifier and zener regulator. we can use this circuit as 12 volt power supply source for stepper motor, servo motor and peripherals need 12V.

Circuit Diagram



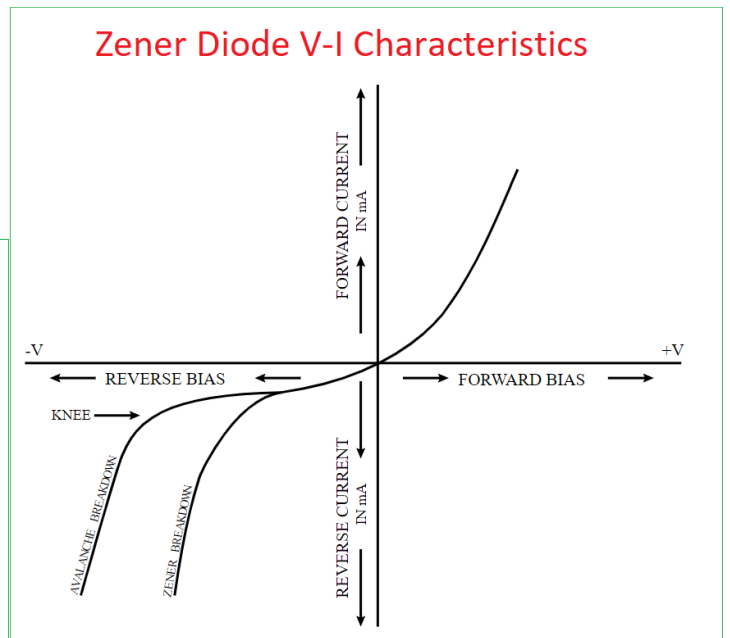
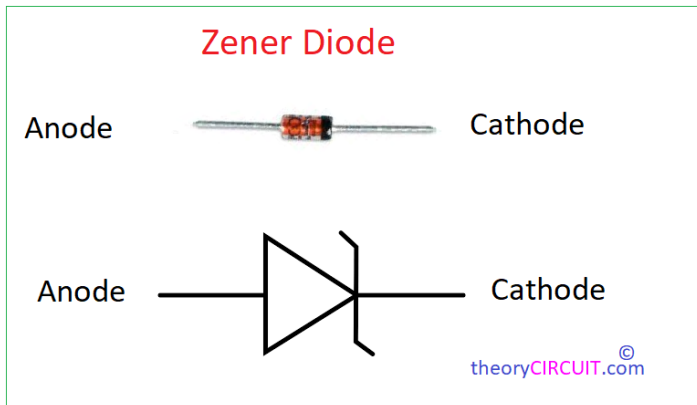
The step down transformer reduce AC supply to 15v AC and four 1N4007 diodes connected as bridge rectifier this will give regulated DC supply. This DC filtered by using C1 (470uF/16v) capacitor and then filtered DC supply fed into 12v zener diode by the way of zener break down the DC supply regulated in 12v. the transistor 2sc1061 drives the output supply to the load.

1.5v Regulated DC Power Supply Circuit using Zener Diode

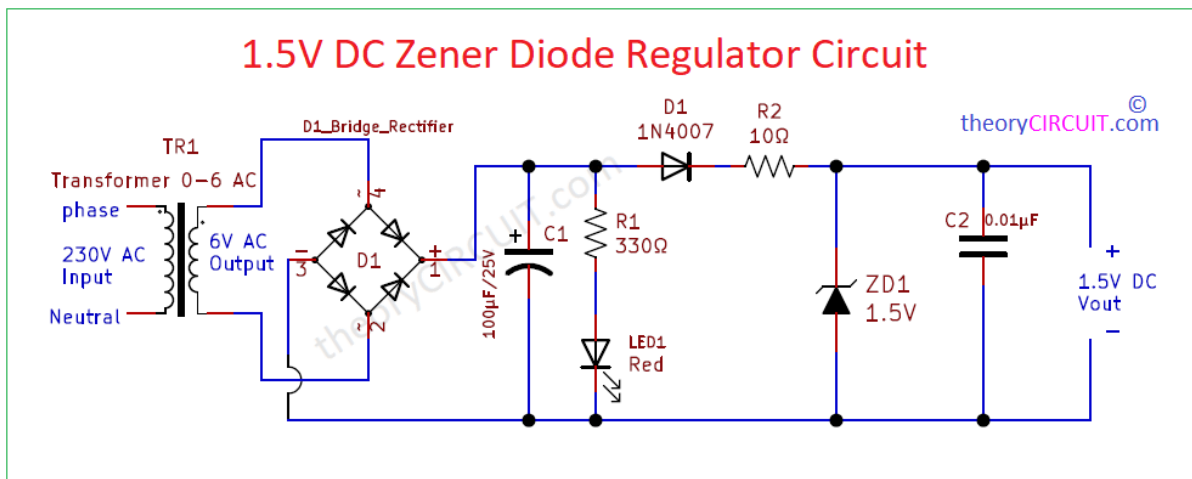
Cost Effective Method of obtaining Regulated DC supply is using Zener Diode as a Voltage Regulator device. In this Article 1.5v Regulated DC Power Supply Circuit using Zener Diode designed and tested along with Bridge Rectifier. In this circuit 1.5V Zener Diode used to Regulate and give 1.5V from unregulated 6V DC supply through bridge Rectifier.

What is Zener Diode?

As we know A Zener Diode or Breakdown diode is a PN diode Specially designed to operate in the breakdown region in Reverse bias condition. Depends on the PN junction doping level this Zener Diode Conducts in Reverse bias when a Certain Specified voltage is reached. This operation called as Zener breakdown.



Circuit Diagram



Components Required (BOM)

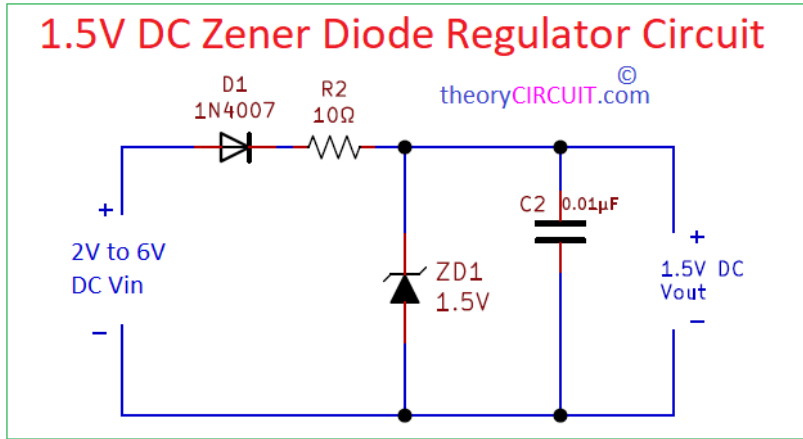
S.No	Designator	Value	Part Number	Quantity
1	TR1	230V to 0-6V AC	0-6V Stepdown Transformer	1
2	C1 C2	100μF/25V 0.01μF	Electrolytic Disc capacitor	1 1
3	R1 R2	330Ω/0.5W 10Ω/0.5W	— —	1 1
4	D1	—	Bridge Rectifier Module	1
5	LED1	Red	—	1

6	ZD1	1.5V	Zener Diode	1
7	D	1N4007	Diode	1

Construction & Working

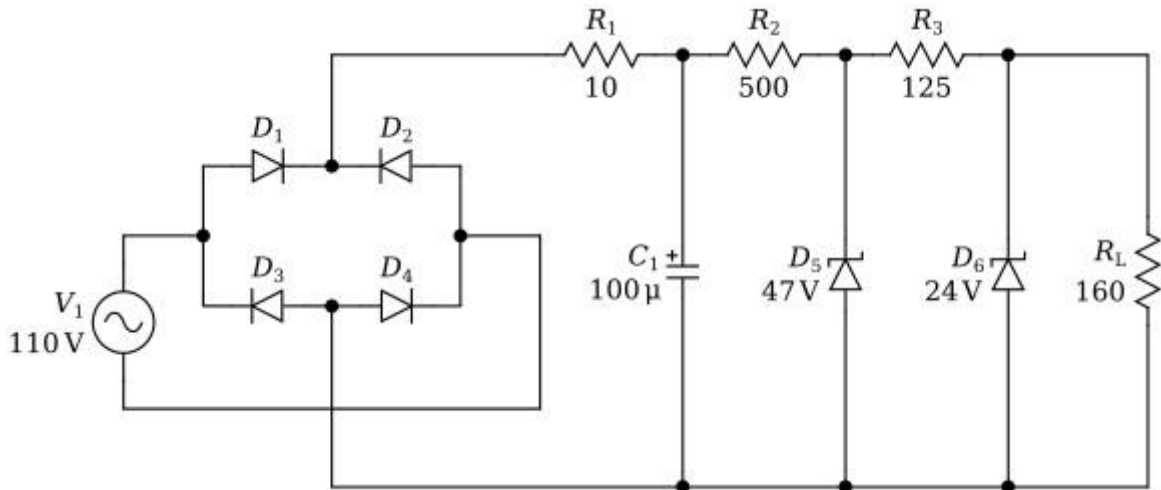
This Circuit Gives Constant Regulated 1.5V DC supply from unregulated DC. This is a standalone power supply circuit hence direct 230V mains supply is reduced to 6V AC by using stepdown transformer, then bridge Rectifier module (or Four 1N4007 Diode in bridge form) Converts 6V AC into 6V DC supply then this DC supply Filtered by Capacitor C1, then LED1 indicates the presence of input DC to the Zener Regulator circuit, Here diode 1N4007 is used as Reverse supply protection device.

1.5V Zener Diode connected parallel and in Reverse bias to the unregulated DC supply. When the Input DC Supply to this Zener diode reaches 1.5V then Zener breakdown occurs and allows only 1.5V at the output end. Capacitor C2 acts as High frequency ripple filter.



If you are using Zener diode in a DC supply Circuit then you can directly connect Zener diode in Reverse bias to obtain Desired Regulated DC supply. Before implementing Zener diode in a Circuit we have to see the Maximum Input and Output Voltage along with current Specifications.

Zener-Diode Regulated Power Supply



This is an example of a two-stage shunt regulator using Zener diodes. The voltage provided at C_1 by the bridge rectifier, comprised of D_1 through D_4 , is about 140 volts (DC) with 10 volts peak to peak ripple. The resistor, R_2 , and the first Zener diode regulator, D_5 , provides additional filtering, reducing the DC voltage to 47 volts. The value of the series resistor, R_2 , was calculated to allow a current of approximately 50 mA (DC) to pass through the first diode, at a nominal input of 120 volts (RMS).

The second regulator provides further ripple reduction while regulating the output at 24 volts for load variations from zero to 150 mA (DC). The output voltage drops about 0.5% when the load is changed from zero to 150 mA (DC).

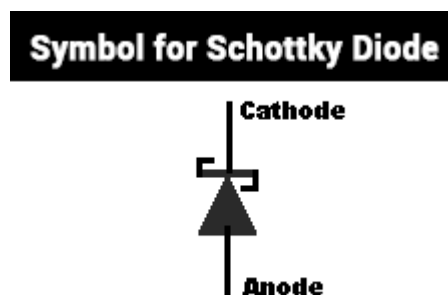
The circuit will take input fluctuations from 100 to 140 volts with little effect on the output. If lower voltages are expected, the circuit may be designed so that diodes will carry a larger current at 155 volts (RMS) input. The upper limit is imposed by the rating of D_5 .

What is Schottky Diode?

The schottky diode is a type of metal – semiconductor junction diode, which is also known as hot-carrier diode, low voltage diode or schottky barrier diode. The schottky diode is formed by the junction of a semiconductor with a metal. Schottky diode offers fast switching action and has a low forward voltage drop. As we are aware that in a PN junction diode, p-type and n-type are joined together to form a PN junction. Whereas, in a Schottky diode metals like platinum or aluminum are used instead of P type semiconductors.

What is Schottky Diode Symbol?

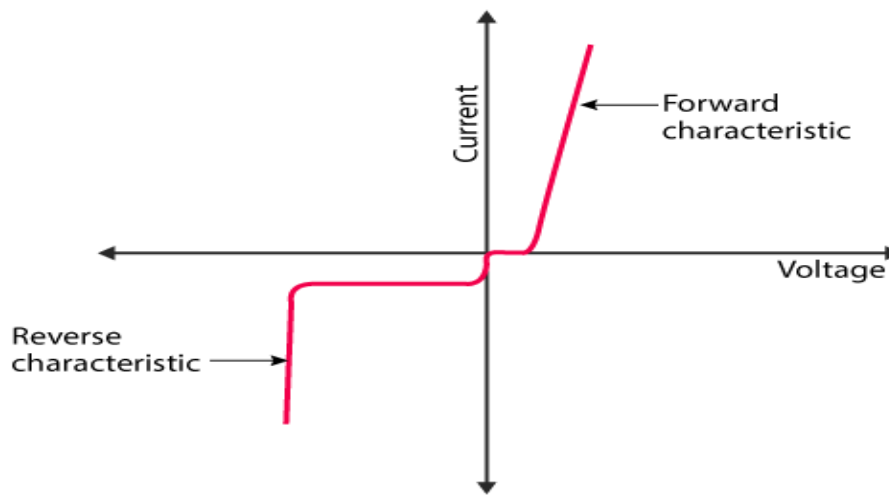
The symbol for the Schottky barrier diode is based around the basic diode circuit symbol. The circuit symbol of the Schottky diode is shown in the figure.



Schottky Diode Symbol

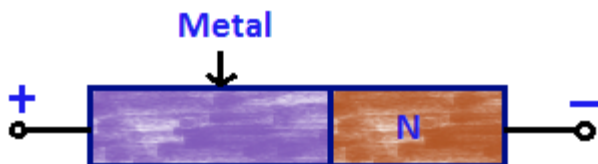
V-I Characteristics of Schottky Diode

The V-I characteristics of Schottky diodes are very much similar to the PN junction diode. Current is the dependent variable while voltage is the independent variable in the Schottky diode. The forward voltage drop of the Schottky diode is low between 0.2 to 0.3 volts.



Working of a Schottky Diode

- The operation relies on the principle that the electrons in different materials have different potential energy.
- N-type semiconductors have higher potential energy than electrons of metals.



- When these two are brought into contact, there is a flow of electrons in both directions across the metal-semiconductor interface.
- A voltage is applied to the Schottky so that the metal is positive when compared to the semiconductor.
- The voltage opposes the built-in potential and makes the current flow easy.

Applications of Schottky Diode

Schottky diodes have been useful for the industry of electronics that has spotted many applications in diode rectifiers because of its unique properties. Here are some major areas where it is widely used.

RF mixer and detector diode:

The Schottky diode consists of its radio frequency functions owing to its switching speed at the highest level and top frequency capability. The Schottky barrier diodes come handy for diode ring mixers with high performance.

Power rectifier:

The Schottky barrier diodes also have functions with high power as rectifiers. The high density of current and voltage drop with low forward shows that the wastage of power is lesser than the normal PN junction diodes.

Power OR circuits:

This diode would be useful for functions where two different power supplies drive a load like in battery supply. It is important that the power coming from supply should not mix with the others.

Solar Cell Applications:

As we know, the solar cells are usually linked to the batteries that are rechargeable, mostly batteries with lead-acid since power supply must be necessary round the clock. Solar cells would not support the applied charge in reverse and thus, a diode would be used in a proportional pattern of the solar cells.

Advantages of Schottky diode

Following are the advantages of Schottky diode:

- The capacitance of the diode is low as the depletion region of the diode is negligible.
- The reverse recovery time of the diode is very fast, that is the change from ON to OFF state is fast.
- The current density of the diode is high as the depletion region is negligible.
- The turn-on voltage of the diode is 0.2 to 0.3 volts, which is very low.

Disadvantages of Schottky diode

The only disadvantage of Schottky diodes is that the reverse saturation current of the diode is large.

What is the difference between Schottky diode and PN junction diode?

Schottky diode	PN junction diode
In this diode, the junction is formed between the n-type semiconductor and the metal plate	In this diode, the junction is formed between the p-type and n-type semiconductors
The forward voltage drop is low	The forward voltage drop for pn junction diode is more
Reverse recovery loss and reverse recovery time are very less	Reverse recovery loss and reverse recovery time are more
It is a unipolar device	It is a bipolar device
The conduction of current happens only due to the movement of electrons	The conduction of current happens due to the movement of electrons and holes

Frequently Asked Questions – FAQs

Why is there no reverse recovery time in the diode?

There are no stored charges as the metal-semiconductor junction is used, due to which the switching is faster.

What are the majority charge carriers in a barrier diode?

Electrons

Why can Schottky diodes be switched off much faster than the p-n junction diode?

As there is no recombination of charges, there is no reverse recovery time which makes switching off faster.

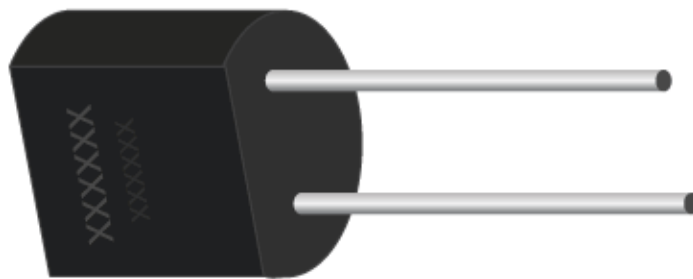
What is reverse recovery time?

Reverse recovery time of the diode is defined as the time taken by the diode to regain its blocking capability.

What is a blocking diode?

A blocking diode is a diode used to direct the energy flow in one direction in a wire.

What is a Varactor Diode?



Varactor Diode

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Varactor Diode

Varactor diode is a type of diode whose internal capacitance varies with respect to the reverse voltage. It always works in reverse bias conditions and is a voltage-dependent semiconductor device. Several names know varactor diode as Varicap, Voltcap, Voltage variable capacitance, or Tuning diode.

Symbol of Varactor Diode

From the diagram given below, it is evident that the symbol of the varactor diode is similar to that of the PN-junction diode. The diode has two terminals: **anode** and **cathode**. One end of the symbol consists of the diode, and the other has two parallel lines representing the capacitor's conductive plates. The gap between the plates shows their dielectric.



Symbol of Varactor Diode

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Operation of Varactor Diode

The function of the varactor diode is to store charges, so it is always operated in reverse bias conditions. When a forward bias voltage is applied, the electric current flows. As a result, the depletion region becomes negligible, which is undesirable.

The junction capacitance of a p-n junction diode is inversely proportional to the width of the depletion layer. In other words, if the width of the depletion layer is less, then the capacitance is more, and vice versa. So if we need to increase the capacitance of a varactor diode, the reverse bias voltage should be decreased. It causes the width of the depletion layer to decrease, resulting in higher capacitance. Similarly, increasing the reverse bias voltage should decrease the capacitance. This ability to get different values of capacitances just by changing the voltage applied is the biggest advantage of a varactor diode compared to a normal variable capacitor.

Varactor Diode Formulae

Capacitance of Varactor Diode	$C_j = CK(V_b - V)^m$
Quality factor of the Varactor Diode	$Q = Ff$

Notations Used In The Varactor Formula

- C_j is the diode capacitance
- C is the diode capacitance when the device is unbiased
- V is the applied voltage
- V_b is the barrier voltage at the junction
- m is the constant depending upon the material
- K is the constant equal to 1
- F is the maximum operating frequency
- f is the operating frequency

Varactor Diode Characteristics

Varactor diodes are mostly operated under reverse bias conditions; therefore, there is no conduction. They are voltage-controlled capacitors and are sometimes known as varicap diodes, although the word varactor is widely used.

Variable Capacitance Characteristics

Variable capacitance effect is shown by normal diodes, but varactor diodes are preferred for providing the required capacitance changes. The diodes are uniquely optimized and manufactured to enable high-range changes in capacitance. Varactor diodes are categorized based on the properties of the diode junction.

Varactor Diode Applications

Some applications of varactor diode are given in the points mentioned below.

- They are used in the RF design arena and provide a method of varying the capacitance within a circuit by applying control voltage. It provides them with special capability, due to which varactor diodes are used in the RF industry.
- These diodes are used in many circuits and find applications in two main sectors.
- FR filters – Tuning filters using varactor diodes is possible. Tracking filters can be required in front-end receiver circuits wherein the filters are enabled to track incoming signals' frequency.
- Voltage-controlled oscillators (VCOs) – VCOs are used for many applications, and oscillators within a phase-locked loop are the major region. VCOs are present in almost all wireless receivers and radios.

Frequently Asked Questions on Varactor Diode

What is abrupt varactor diode?

The abrupt varactor diode exhibits the inverse square of the C-V function, and the doping concentration is constant.

What is a hyper abrupt varactor diode?

A Hyperabrupt varactor diode is used as it has greater capacitance for the given voltage change.

Which layer can be used for the detection of bad frames?

The link layer is used for the detection of bad frames.

What is the major use of varactor diode?

The major use of the varactor diode is for tuning the circuit.

What happens to plate separation when the depletion region in the varactor diode widens?

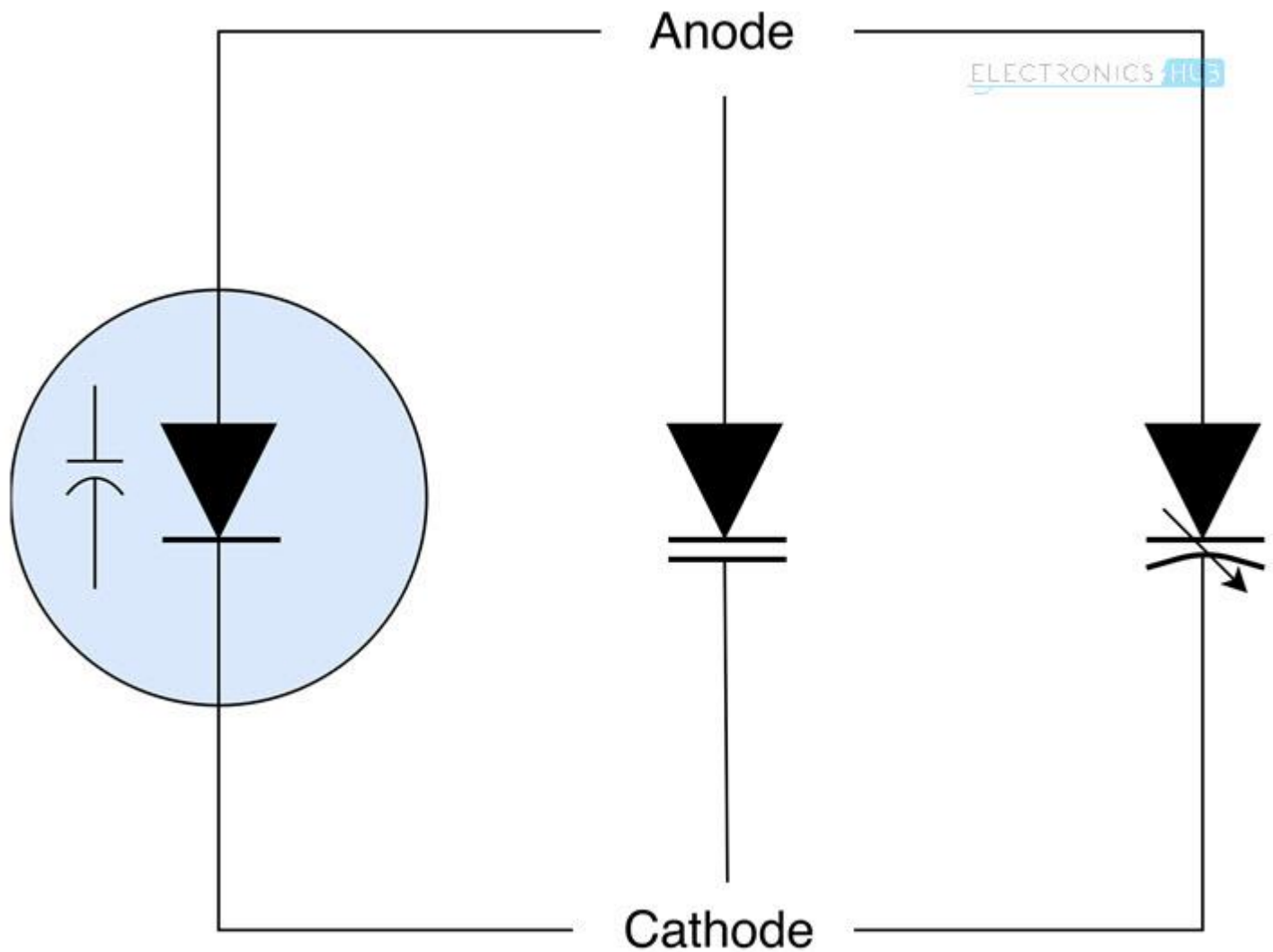
As the depletion region in the varactor diode widens, the plate separation increases.

Which type of diode is commonly used in electronic tuners in TVs?

Varactor diode is the commonly used diode in electronic tuners in TVs.

Symbolic Representation

A number of schematic symbols are used to represent a Varactor Diode in circuit diagrams. Of them, the three most popular representations are shown below:

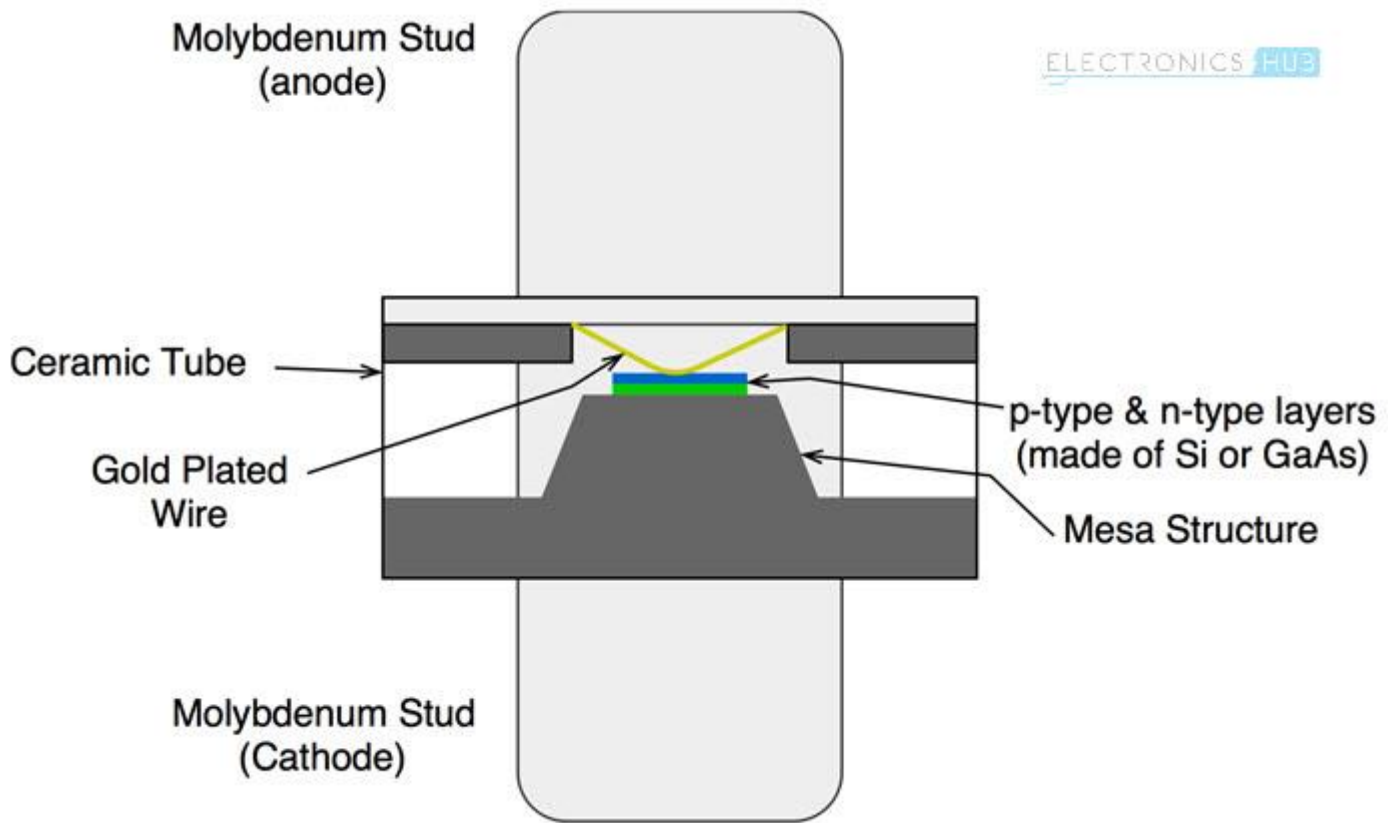


Among these three symbols of a Varactor Diode, the second symbol is more frequently used. It is a combination of symbol of a p-n junction diode and a capacitor. In the symbol, the triangle indicates the presence of a diode component, and the two lines at the tip of triangle indicate parallel plates of a capacitor.

Construction

A Varactor Diode consists of p-type and n-type semiconductor layers sandwiched together, with the n-type layer attached to a mesa (table-shaped) structure. A gold plated molybdenum stud is connected to n-type layer via the mesa structure and it acts as cathode terminal.

The p-type layer is connected to another gold plated molybdenum stud (which acts as anode) via a gold wire. Except for some portion of the molybdenum studs, the entire arrangement is enclosed in a ceramic layer.



The p-type and n-type layers of the varactor diode are made up of silicon or gallium arsenide depending on the type of application for which it is used. For low frequency applications, silicon is used, and for high-frequency applications gallium arsenide is used.

For conventional diodes, the p-type and n-type semiconductor layers are uniformly doped with impurities to improve conductivity. But in the case of varactor diodes, the concentration of impurities near the pn junction is very less and it gradually increases as we move towards the layer's other surface.

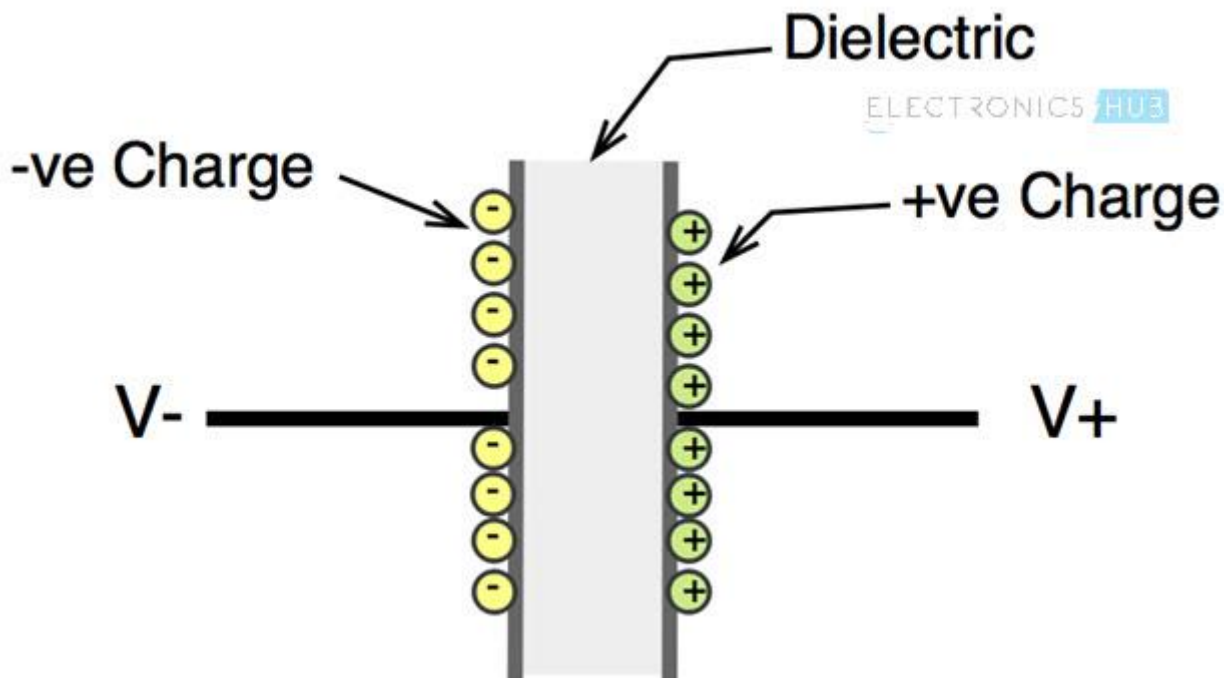
Working

To learn how a varactor diode works, you need to first understand the working principle of a variable capacitor:

A capacitor consists of two conducting surfaces separated by a non-conducting dielectric medium (see figure below). When one of surfaces is connected to a positive voltage and the other to negative voltage, because of attraction between positive and negative carriers, positive charge accumulates on one surface and negative charge on the other.

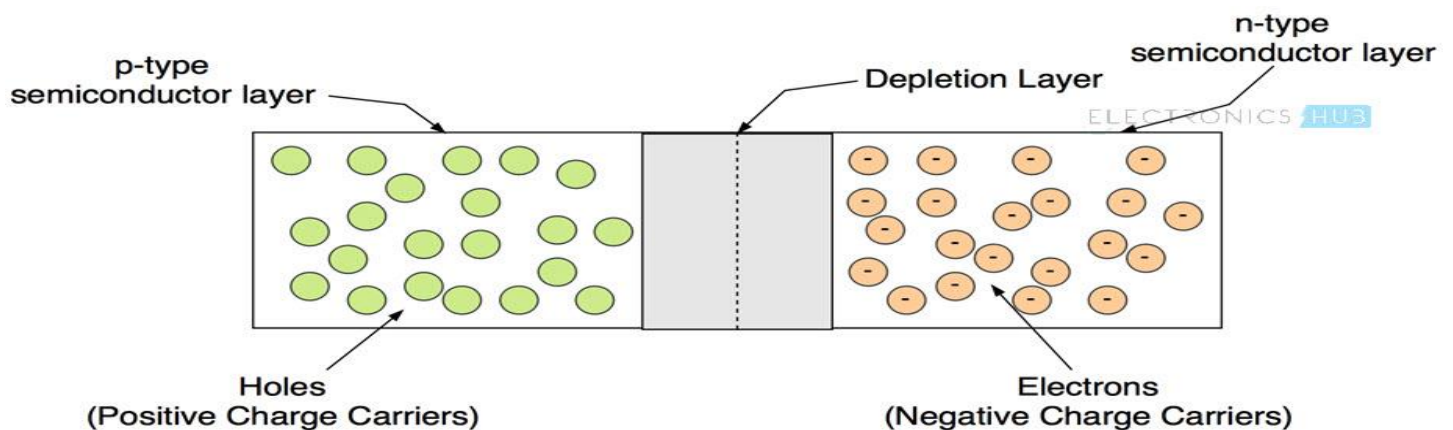
The amount of charge that accumulates is termed as capacitance. If we reduce the gap between the two surfaces, the force of attraction between positive and negative charge carriers increases and so more charge accumulates on the surface i.e. the capacitance increases.

The opposite happens when the surfaces are moved away from each other i.e. the capacitance decreases. A variable capacitor has a mechanical arrangement which allows us to change the gap between surfaces, which effectively changes the capacitance.



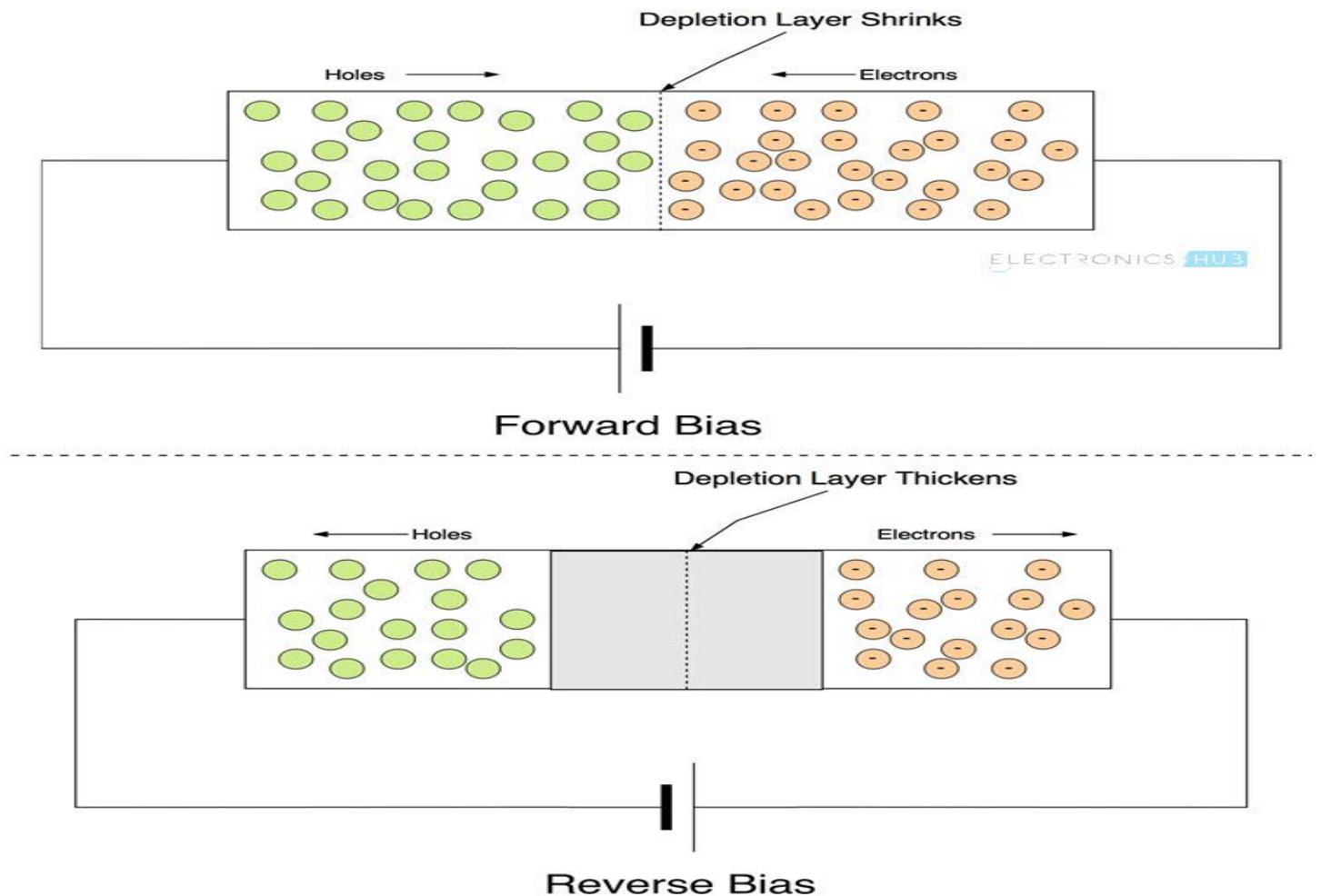
Let us now come back to working of a p-n junction diode. The p-type layer of the diode is filled with positive charge carriers and the n-type layer with negative charge carriers. Near the surface of contact between the two layers, positive and negative charges diffuse and neutralize each other. This region is known as depletion region.

No further diffusion of charge carriers through the depletion layer is possible unless an external voltage is applied. So effectively the depletion layer acts as an insulator.



The width of depletion layer depends on the voltage applied across the p and n-type layers. If a forward bias voltage is applied, i.e. positive voltage is applied to the p-type layer and negative voltage is applied to the n-type layer, the width of depletion decreases, and above a certain voltage it completely disappears.

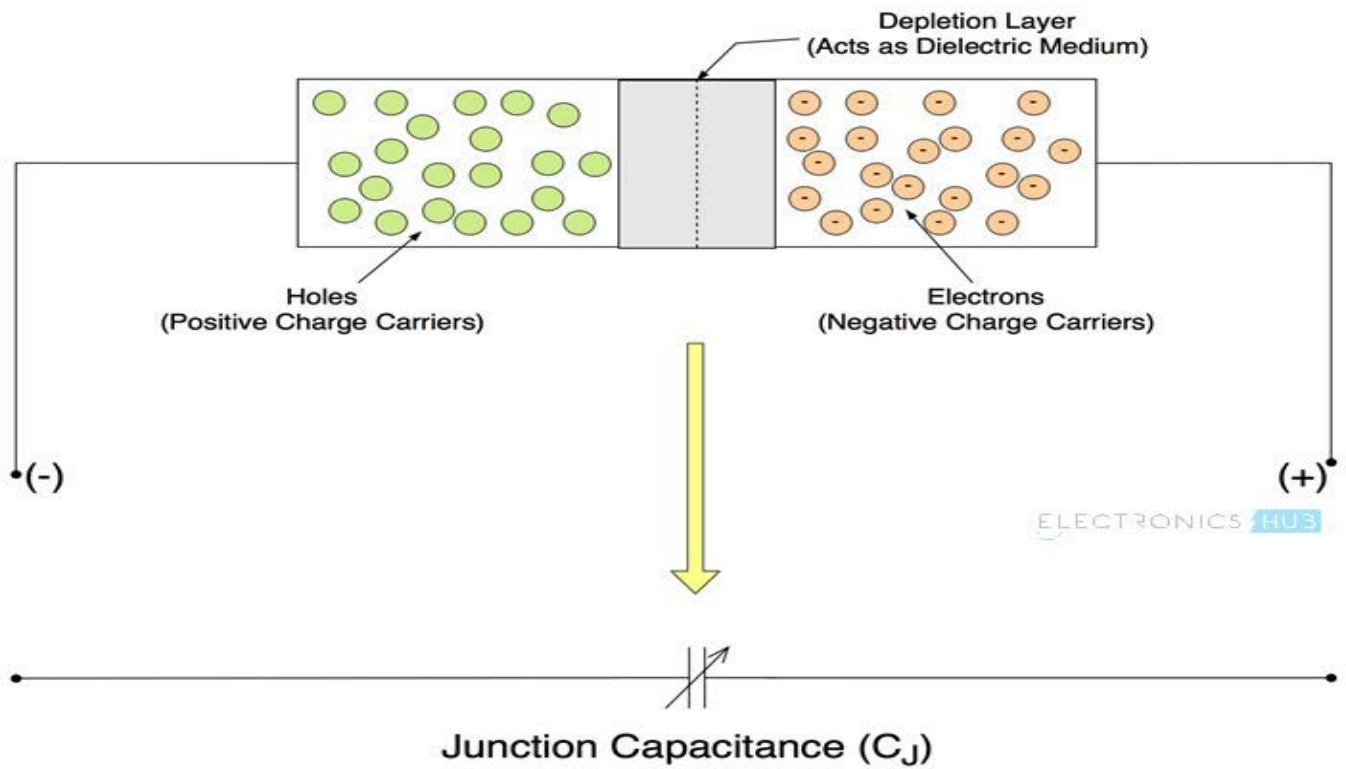
If a reverse bias is applied, i.e., positive voltage to n-type layer and negative voltage to p-type layer, the width of depletion layer increases. The below diagram illustrates the two scenarios:



Simply put, the width of the depletion region can be changed to a required value just by adjusting the voltage across the p & n-type semiconductor layers. So, by now you must have observed the similarities between capacitor, and the diode in reverse bias. The depletion layer in the diode is akin to the dielectric medium in a capacitor, which acts as an insulator and prevents the flow of charge carriers from one side to the other.

So, when reverse bias voltage is applied across the diode, respective charge carriers accumulate on either side of the depletion layer. This makes the diode acquire some capacitance and it is termed as junction capacitance.

A Varactor Diode is specially designed to enhance this ability to store charge carriers when reverse bias is applied, thus allowing it to act as a capacitor.

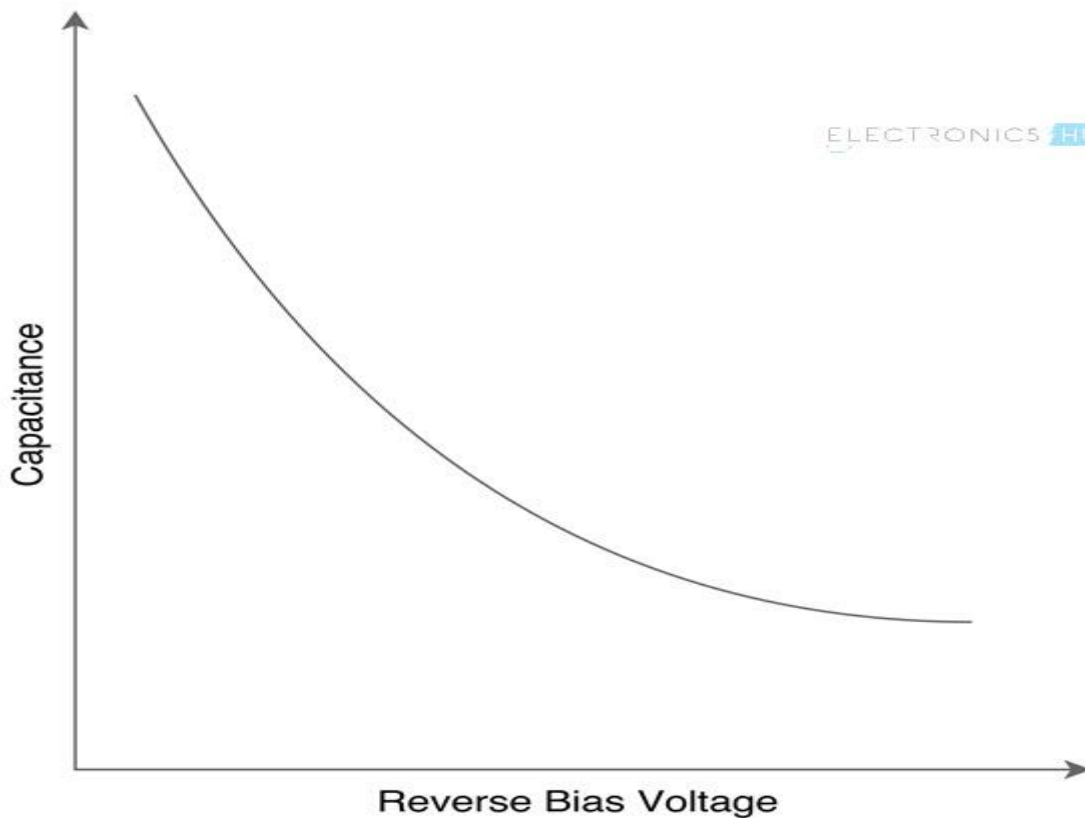


The junction capacitance is inversely proportional to the width of depletion layer i.e. if the width of depletion layer is less, the capacitance will be more, and vice versa. So if we need to increase the capacitance of a varactor diode, the reverse bias voltage should be decreased. It causes the width of depletion layer to decrease, resulting in higher capacitance. Similarly increasing the reverse bias voltage should decrease the capacitance.

This ability to get different values of capacitances just by changing the voltage applied is the biggest advantage of a varactor diode when compared with a normal variable capacitor.

Characteristics

The below graph illustrates the relation between the magnitude of reverse bias voltage p-type & n-type layers of a varactor diode, and the magnitude of junction capacitance.



You can observe that the junction capacitance of a varactor diode is inversely proportional to reverse bias voltage. Also because of difference in the way impurities are added to the p-type and n-type layers, the capacitance of a varactor diode is always higher than that of a conventional diode.

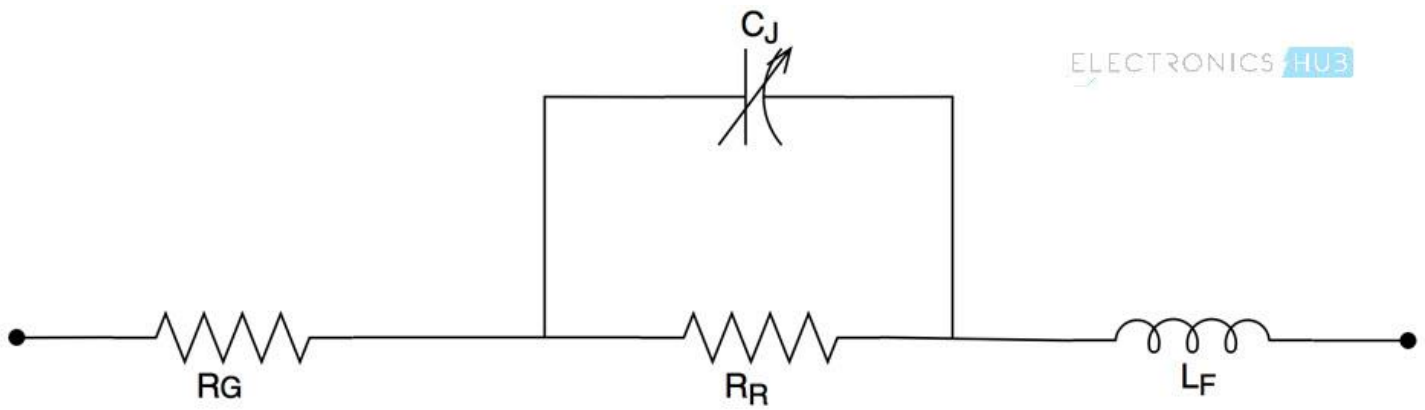
The exact magnitude of junction capacitance of a varactor diode is calculated with respect to the capacitance at zero-bias condition (C_J), forward bias voltage (Voltage required to remove depletion layer completely, V_B) and the actual reverse bias voltage (V_R) applied across the junction. It is given by the formula:

$$C_J = C_0(1 + V_R/V_B)^{-n}$$

‘n’ is a constant and varies depending on the way p and n-type layers are doped. Its value lies between (0.5 – 0.33). The value of reverse bias voltage (V_R) should be below the breakdown voltage, above which the value of capacitance C_J equals zero because of breakdown of depletion region and the free flow of holes and electrons.

Equivalent Circuit

For designing circuits that use the properties of a varactor diode, we first need to break down its electrical properties and visualize an equivalent circuit with basic components like resistors, capacitors and inductors. The below image shows such approximate equivalent circuit of a varactor diode when operated under reverse bias condition.



R_R is the reverse bias resistance and R_G is the geometric resistance of the varactor diode. C_J is the junction capacitance which is dependent on the reverse bias voltage. L_F is the effective inductance and it depends on the frequency limits under which a varactor diode should be operated.

Applications

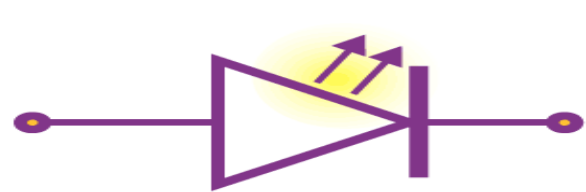
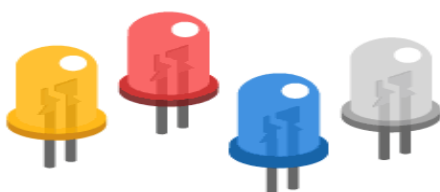
Owing to the special property of varying capacitance with varying voltage, varactor diodes are mostly used in frequency modulation or tuning circuits where the value of capacitance determines the output modulation frequency. Some of the other applications include:

- Automatic Frequency Controllers (AFCs)
- Ultra High Frequency Television sets
- High frequency Radios
- Frequency Multipliers
- Band Pass Filters
- Harmonic Generators.

What is LED?

A light-emitting diode (LED) is a semiconductor device that emits light when an electric current flows through it. When current passes through an LED, the electrons recombine with holes emitting light in the process. LEDs allow the current to flow in the forward direction and blocks the current in the reverse direction.

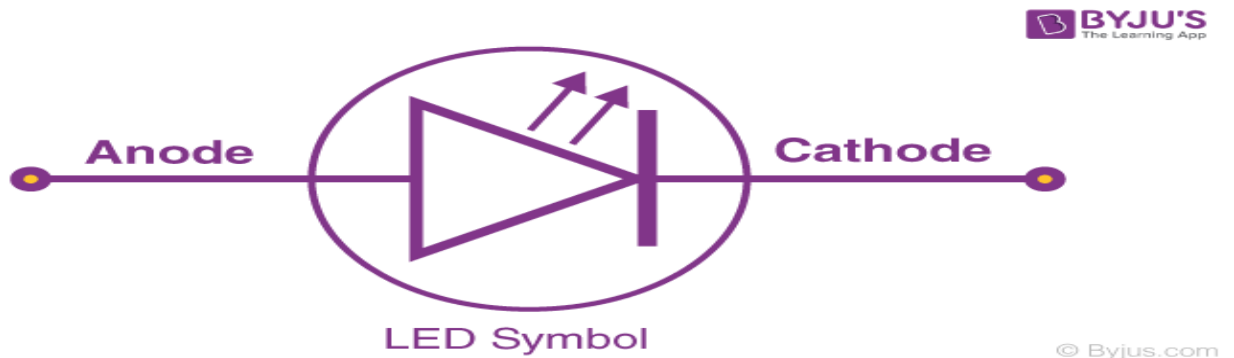
LIGHT EMITTING DIODES



Light-emitting diodes are heavily doped p-n junctions. Based on the semiconductor material used and the amount of doping, an LED will emit coloured light at a particular spectral wavelength when forward biased. As shown in the figure, an LED is encapsulated with a transparent cover so that emitted light can come out.

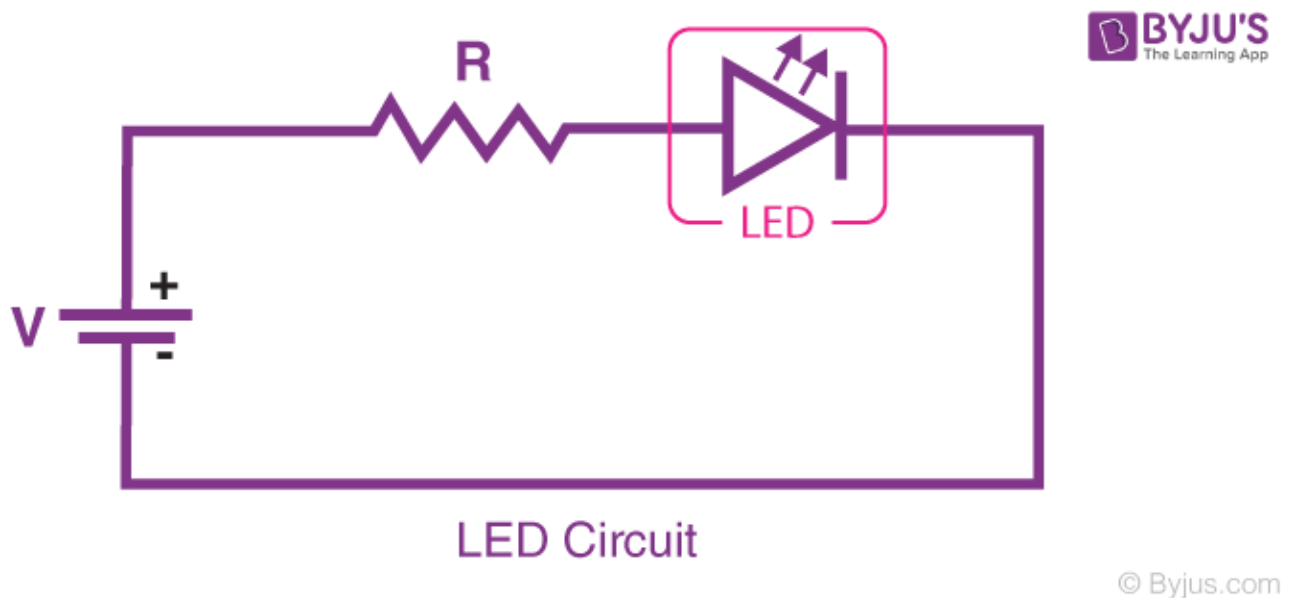
LED Symbol

The LED symbol is the standard symbol for a diode, with the addition of two small arrows denoting the emission of light.



Simple LED Circuit

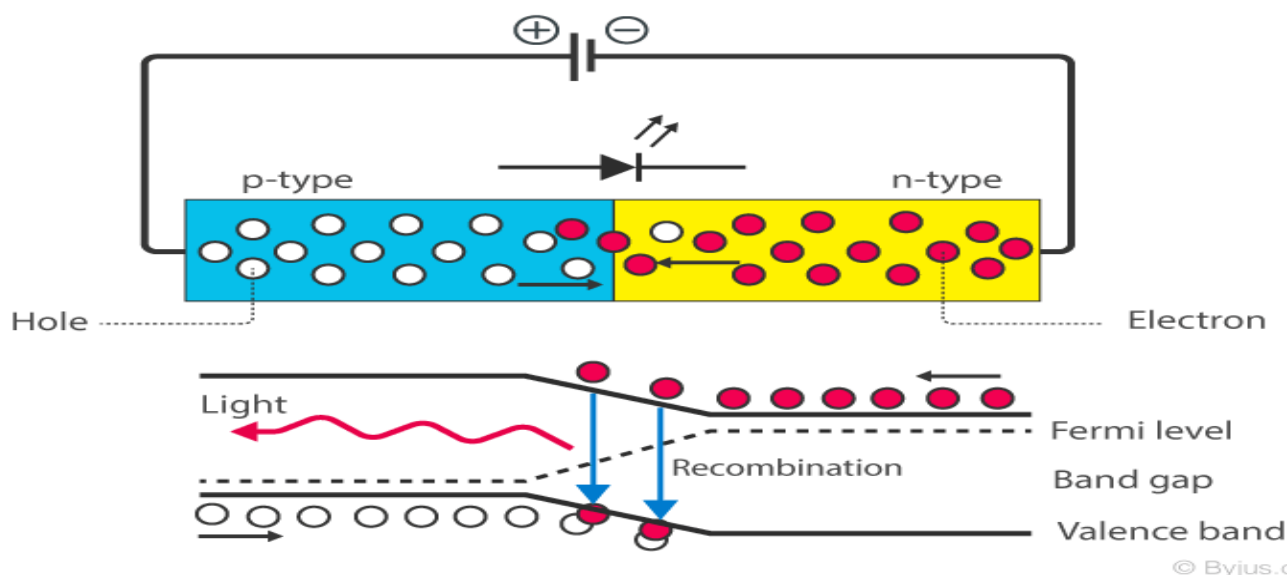
The figure below shows a simple LED circuit.



The circuit consists of an LED, a voltage supply and a resistor to regulate the current and voltage.

How does an LED work?

When the diode is forward biased, the minority electrons are sent from $p \rightarrow n$ while the minority holes are sent from $n \rightarrow p$. At the junction boundary, the concentration of minority carriers increases. The excess minority carriers at the junction recombine with the majority charges carriers.



The energy is released in the form of photons on recombination. In standard diodes, the energy is released in the form of heat. But in light-emitting diodes, the energy is released in the form of photons. We call this phenomenon electroluminescence. Electroluminescence is an optical phenomenon, and electrical phenomenon where a material emits light in response to an electric current passed through it. As the forward voltage increases, the intensity of the light increases and reaches a maximum.

What determines the colour of an LED?

The colour of an LED is determined by the material used in the semiconducting element. The two primary materials used in LEDs are aluminium gallium indium phosphide alloys and indium gallium nitride alloys. Aluminium alloys are used to obtain red, orange and yellow light, and indium alloys are used to get green, blue and white light. Slight changes in the composition of these alloys change the colour of the emitted light.

Properties of Laser Light

Laser light is monochromatic, directional and coherent.

Laser Light is Monochromatic

Unlike white light, which is made of seven colours, laser light is made of a single colour.

Laser Light is Directional

Laser light is highly directional.

Laser Light is Coherent

Laser light is coherent because the wavelengths of the laser light are in phase in space and time.

Uses of LED

LEDs find applications in various fields, including optical communication, alarm and security systems, remote-controlled operations, robotics, etc. It finds usage in many areas because of its long-lasting capability, low power requirements, swift response time, and fast switching capabilities. Below are a few standards LED uses:

- Used for TV back-lighting
- Used in displays

- Used in Automotives
- LEDs used in the dimming of lights

Types of LED

Below is the list of different types of LED that are designed using semiconductors:

- Miniature LEDs
- High-Power LEDs
- Flash LED
- Bi and Tri-Colour
- Red Green Blue LEDs
- Alphanumeric LED
- Lighting LED

Advantages of LEDs over Incandescent Power Lamps

Some advantages of LEDs over Incandescent Power Lamps are:

- LEDs consume less power, and they require low operational voltage.
- No warm-up time is needed for LEDs.
- The emitted light is monochromatic.
- They exhibit long life and ruggedness.

Read More: [Types of LED](#)

Frequently Asked Questions – FAQs

What is LED?

A light-emitting diode (LED) is a semiconductor device that emits light when an electric current flows through it.

What is Light Emitting Diodes used for?

LEDs have a wide range of applications ranging from your mobile phone to large advertising billboards. They mostly find applications in devices that show the time and display different types of data.

How do LEDs work?

LEDs work on the principle of Electroluminescence. On passing a current through the diode, minority charge carriers and majority charge carriers recombine at the junction. On recombination, energy is released in the form of photons. As the forward voltage increases, the intensity of the light increases and reaches a maximum.

What is Electroluminescence?

Electroluminescence is an optical phenomenon, and electrical phenomenon where a material emits light in response to an electric current passed through it.

What are the advantages of LEDs?

LEDs consume less power, and they require low operational voltage. No warm-up time is needed for LEDs.

The uses and functions of LEDs depend upon the place where they are used, some examples are mentioned below:

LEDs used for TV Backlighting

A TV's backlight is the major power-consuming source. Uses of LEDs can give an efficient power reduction. On the edges of the TV, using an LED will be a cost reduction choice. Using LEDs directly behind the display provides better contrast. When it comes to TV backlighting, LEDs have taken the place of CFLs and LCDs.

It is used for Smartphone Backlighting

The backlight design of the smartphone may be slimmer and less expensive thanks to the usage of LED. The price of LEDs may vary according to the size of the smartphone display. They provide greater battery life due to the lower output voltage.

Uses of LED in Displays

LED display boards are common these days and are used outdoors like storage signs, billboards, road signs, etc. In sign boards, which have multiple languages conveying signals, the use of more LEDs will be beneficial in terms of less power consumption.

These are used in Automotives

The Use of LEDs in the automotive industry is growing. With LEDs, energy is saved and there is clearer visibility. These are extensively used in the back and rear of an automobile for better accessibility. LED lighting can improve the safety of pedestrians and drivers as it enhances visibility when it is ON, OFF, and dimmed in any part of the journey.

LEDs used in Dimming of Lights

Few LED applications include dimming of lights which helps in reducing energy consumption.

This dimming feature is also used in Appliances where it is of two types.

Global Dimming where all LEDs are dimmed together.

Local Dimming where LEDs are dimmed independently.

These were some applications of LEDs if you wish to know more, download BYJU'S The Learning App.

Frequently Asked Questions – FAQs

What is meant by a semiconductor?

A semiconductor is a material that possesses electrical conductivity between an insulator and a conductor. Semiconductors are used in the production of different types of electronic devices, including transistors, integrated circuits and diodes.

What is a diode?

A diode is an electrical device that allows the flow of charges (current) in only one direction. A diode is denoted by a triangle symbol with a single line across one vertex.

What is the full form of an LED?

Light-emitting diode is the full form of LED.

What are the common uses of LEDs?

LEDs are used in TV backlighting, smartphone backlighting, LED displays, automotive lighting, etc.

Give a few examples of LED displays.

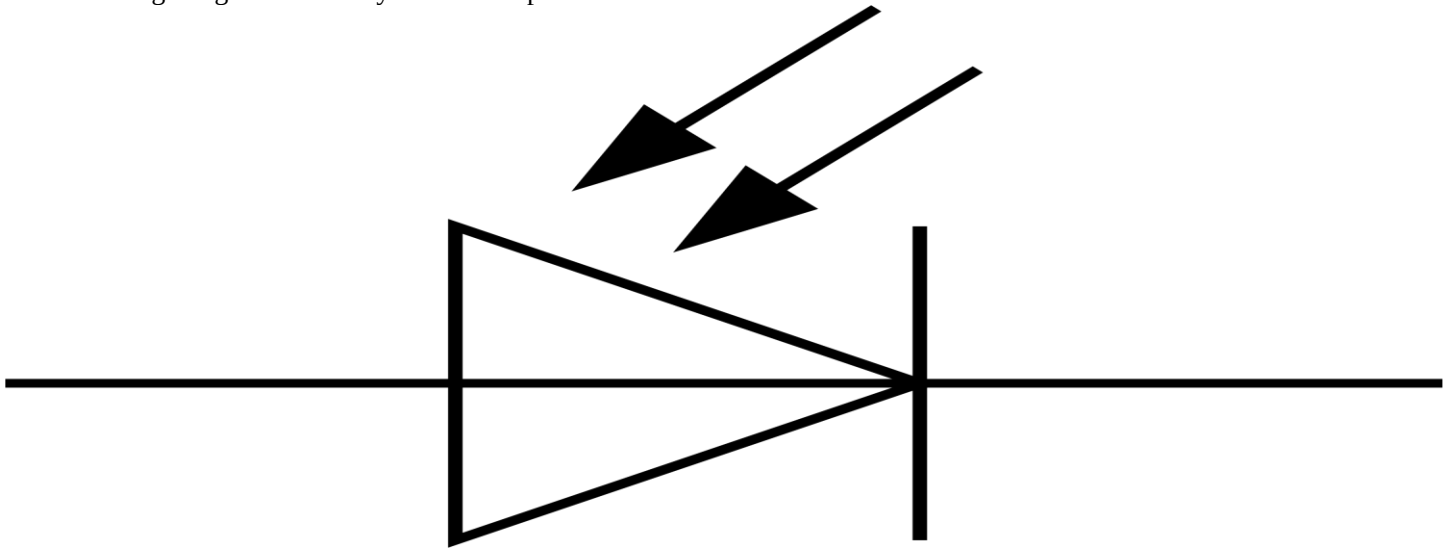
What is Photodiode?

A photodiode is a PN-junction diode that consumes light energy to produce an electric current. They are also called a photo-detector, a light detector, and a photo-sensor. Photodiodes are designed to work in reverse bias condition. Typical photodiode materials are Silicon, Germanium and Indium gallium arsenide.



Symbol of Photodiode

The following image shows the symbol of the photodiode:



The symbol of the photodiode is similar to that of an LED, but here the arrow points inwards.

Photodiode Working

A photodiode is subjected to photons in the form of light which affects the generation of electron-hole pairs. If the energy of the falling photons ($h\nu$) is greater than the energy gap (E_g) of the semiconductor material, electron-hole pairs are created near the depletion region of the diode. The electron-hole pairs created are separated from each other before recombining due to the electric field of the junction. The direction of the electric field in the diode forces the electrons to move towards the n-side and consequently the holes move towards the p-side. As a result of the increase in the number of electrons on the n-side and holes on the p-side, a rise in the electromotive force is observed. Now when an external load is connected to the system, a current flow is observed through it.

The more the electromotive force created, the greater the current flow. The magnitude of the electromotive force created depends directly upon the intensity of the incident light. This effect of the proportional change in photocurrent with the change in light intensity can be easily observed by applying a reverse bias.

Since photodiodes generate current flow directly depending upon the light intensity received, they can be used as photodetectors to detect optical signals. Built-in lenses and optical filters may be used to enhance the power and productivity of a photodiode.

Applications of Photodiode

- Photodiodes are used in simple day-to-day applications. The reason for their prominent use is their linear response of photodiode to light illumination.
- Photodiodes with the help of optocouplers provide electric isolation. When two isolated circuits are illuminated by light, optocouplers are used to couple the circuit optically. Optocouplers are faster compared to conventional devices.
- Photodiodes are used in safety electronics such as fire and smoke detectors.
- Photodiodes are used in numerous medical applications. They are used in instruments that analyze samples, detectors for computed tomography and also used in blood gas monitors.
- Photodiodes are used in solar cell panels.
- Photodiodes are used in logic circuits.
- Photodiodes are used in the detection circuits.
- Photodiodes are used in character recognition circuits.
- Photodiodes are used for the exact measurement of the intensity of light in science and industry.
- Photodiodes are faster and more complex than normal PN junction diodes and hence are frequently used for lighting regulation and optical communication.

Stay tuned to BYJU'S to know more about the modern-day uses of photodiodes with interesting video lectures and fall in love with learning!

Frequently Asked Questions – FAQs

What is a photodiode?

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Write some applications of photodiodes.

Photodiodes are used in logic circuits, solar cell panels, detection circuits, lighting regulation, optical communication etc.

What is electromotive force?

Electromotive force is defined as the electric potential produced by either an electrochemical cell or by changing the magnetic field. EMF is the commonly used acronym for electromotive force.

What kind of metals exhibit the photoelectric effect?

Alkali metal is usually used for exhibiting a photoelectric effect. Metals such as copper and sodium are the examples of metals that exhibit photoelectric effect.

What kind of light is needed to cause the photoelectric effect?

Ultraviolet light is the type of light that is needed to cause the photoelectric effect.

What type of bias is used for photodiodes?

A photodiode always operates under reverse bias condition.

Is photodiode a sensor?

Yes, a photodiode can be used as a light sensor and convert light energy into electrical energy.

What is the effect of light on photodiodes?

When the diode is illuminated, photons of light are absorbed generating electron-hole pairs.